

STAT 216 Coursepack



Fall 2026
Montana State University

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Preface

This coursepack accompanies the textbook for STAT 216: Montana State Introductory Statistics with R, which can be found at <https://mtstateintrostats.github.io/IntroStatTextbook/>. The syllabus for the course (including the course calendar), data sets, and links to Canvas, Gradescope, and the MSU RStudio server can be found on the course webpage: <https://math.montana.edu/courses/s216/>. Other notes and review materials are linked in Canvas.

Each of the activities in this workbook is designed to target specific learning outcomes of the course, giving you practice with important statistical concepts in a group setting with instructor guidance. In addition to the in-class activities for the course, video notes are provided to aid in taking notes while you complete the required videos. Bring this workbook with you to class each class period, and take notes in the workbook as you would your own notes. A well-written completed workbook will provide an optimal study guide for exams!

All activities and labs in this coursepack will be completed during class time. Parts of each lab will be turned in on Gradescope. To aid in your understanding, read through the introduction for each activity before attending class each day.

Additional review activities for each Module are provided at the end of each Unit. These review activities will be completed during weekly review sessions throughout the semester.

STAT 216 is a 3-credit in-person course. In our experience, it takes six to nine hours per week outside of class to achieve a good grade in this class. By “good” we mean at least a C because a grade of D or below does not count toward fulfilling degree requirements. Many of you set your goals higher than just getting a C, and we fully support that. You need roughly nine hours per week to review past activities, read feedback on previous assignments, complete current assignments, and prepare for the next day’s class. A typical week in the life of a STAT 216 student looks like:

- *Prior to class meeting:*
 - Read assigned sections of the textbook, using the provided reading guides to take notes on the material.
 - Watch the provided videos, taking notes in the coursepack.
 - Read through the introduction to the day’s in-class activity.
 - Read through the week’s homework assignment and note any questions you may have on the content.
- *During class meeting:*
 - Work through the activities and labs with your classmates and instructor, taking detailed notes on your answers to each question in the activity.
- *After class meeting:*
 - Complete any parts of the activity you did not complete in class.
 - Review the activity solutions in the Math and Stat Center, and take notes on key points.
 - Complete any remaining assigned readings videos, and quizzes for the week.
 - Complete the week’s homework assignment.
 - Attend weekly review sessions or complete the review worksheets

Basics of Data and Sampling Methods

1.1 Vocabulary Review and Key Topics

At the beginning of each module is a summary of key topics and new vocabulary terms for that module. As you read through the material in the textbook and watch the videos prior to class, look for these terms. Reference the definitions to guide your understanding.

1.1.1 Key topics

Module 1 introduces the foundations of data: observational units, types of variables, and how to collect sample data from a population of interest in a way that allows us to generalize our results back to the population.

1.1.2 Vocabulary

- **Data:** observations used to answer research questions.
- **Observational units (cases):** the subjects or entities on which data are collected.
 - The rows in a data set represent the observational units.
- **Sample size:** the number of observational units in a data set, denoted by n .
- **Variable:** the characteristics collected on each observational unit.
- **Types of variables:**
 - **Categorical:** cases are grouped into categories.
 - **Quantitative:** numerical measurements, where performing arithmetic operations makes sense.
- **Target population:** group of observational units of interest.
- **Sample:** subset of the population.
- **Statistic:** numerical value calculated on a sample; for categorical data, a proportion is calculated, for quantitative data, the mean is calculated.
- **Parameter:** numerical value of the entire population we are interested in; this is generally an unknown value.
- **Sampling methods:**
 - **Unbiased sampling method (e.g., a random sample):** on average, the sample will be representative of the target population; all observational units in the target population have the same chance of being selected.
 - **Biased sampling method (e.g., convenience sample):** on average, the sample will not be representative of the target population; some part of the target population will be over- or under-represented.
- **Type of sampling bias:**
 - **Selection bias:** method of sampling is biased; some part of the target population is over- or under-represented.

- **Non-response bias:** part of a pre-selected sample does not respond or cannot be reached.
- **Response bias:** responses are not truthful (poor/leading question phrasing, social desirability).
- **Generalization:** to what group of observational units can the results be applied to?
 - If an unbiased method of selection was used and there is no non-response or response bias, we can generalize the results to the target population.
 - If a biased method of selection was used or if non-response or response bias is present, we can only generalize the result to the sample or similar observational units.

1.2 Video Notes: Intro to data and Sampling Methods

Read through Sections 1.1 – 1.3 and 2.1 in the course textbook and watch the course videos prior to coming to class. Fill in the following questions to aid in your understanding of the material.

1.2.1 Course Videos

- Data_Basics
- Sampling

Video: Data Basics (sections 1.2.1 and 1.2.2)

Data: observations used to answer research questions

Observational unit or case: the people or things we _____ data from; in a data set, each _____ represents one observational unit.

- The number of observational units in a data set is called the _____
_____ and is denoted by the letter _____ .

Variable: _____ measured on each observational unit; in a dataset, each _____ represents one variable.

Types of variables

- Categorical variable:

- Quantitative variable:

Example: Review the `loan50` data set discussed in section 1.2.1 of the textbook.

- Identify the sample size. Write it using proper notation.

- What are the observational units?

- Review Table 1.4 in section 1.2.1 of the textbook, listing the 7 variables in the `loan50` data set. Identify which variables are categorical.

- Identify which variables are quantitative.

Summary measure: the *type* of number (mean/proportion/correlation/etc.) used to summarize an entire data set

Examples:

proportion of loans when the person owns a home

mean (or average) interest rate

- The summary measure (mean/proportion/correlation/etc.) and type of plot (bar plot/histogram/scatterplot/etc.) used depends on the type (categorical or quantitative) of variable(s) being summarized!

Optional Notes: Additional Example

The Bureau of Transportation Statistics (*Bureau of Transportation Statistics* 2019) collects data on all forms of public transportation. The data set seen here includes several variables collect on flights departing on a random sample of 150 US airports in December of 2019.

```
airport <- read.csv("data/airport_delay.csv")
glimpse(airport)
#> Rows: 150
#> Columns: 19
#> $ airport      <chr> "ABI", "ABY", "ACV", "ACY", "ADQ", "AEX", "ALB", "~
#> $ city         <chr> "Abilene", "Albany", "Arcata/Eureka", "Atlantic Ci~
#> $ state        <chr> " TX", " GA", " CA", " NJ", " AK", " LA", " NY", "~
#> $ airport_name <chr> " Abilene Regional", " Southwest Georgia Regional"~
#> $ hub          <chr> "no", "no", "no", "no", "no", "no", "no", "no", "n~
#> $ international <chr> "no", "no", "no", "yes", "no", "yes", "yes", "yes"~
#> $ elevation_1000 <dbl> 1.7906, 0.1932, 0.2223, 0.0748, 0.0787, 0.0881, 0.~
#> $ latitude     <dbl> 32.4, 31.5, 41.0, 39.5, 57.7, 31.3, 42.7, 35.2, 45~
#> $ longitude    <dbl> -99.7, -81.2, -124.1, -74.6, -152.5, -92.5, -73.8,~
#> $ arr_flights  <int> 195, 81, 215, 293, 54, 282, 943, 410, 53, 32314, 6~
#> $ perc_delay15 <dbl> 16.410256, 13.580247, 23.255814, 15.358362, 12.962~
#> $ perc_cancelled <dbl> 0.5128205, 0.0000000, 4.1860465, 0.6825939, 14.814~
#> $ perc_diverted <dbl> 0.00000000, 0.00000000, 2.32558139, 0.68259386, 0.~
#> $ arr_delay    <int> 1563, 1244, 4763, 2905, 329, 1293, 15127, 9705, 25~
#> $ carrier_delay <int> 459, 890, 1613, 476, 180, 302, 5627, 2253, 439, 10~
#> $ weather_delay <int> 21, 43, 549, 124, 1, 58, 2346, 168, 1236, 13331, 2~
#> $ nas_delay     <int> 257, 39, 154, 771, 51, 112, 2096, 616, 746, 45674,~
#> $ security_delay <int> 0, 0, 0, 25, 0, 0, 44, 0, 0, 375, 0, 83, 0, 23, 0,~
#> $ late_aircraft_delay <int> 826, 272, 2447, 1509, 97, 821, 5014, 6668, 108, 10~
```

- What are the observational units?
- Identify which variables are categorical.
- Identify which variables are quantitative.

Video: Sampling (section 2.1)

The method used to collect data will impact **who the results apply to** and **what type of conclusions** can be drawn.

- Target population: all _____ of interest
- Sample: the group of _____ from which data is collected; a subset of the target population

Parameters and statistics

Statistic: a single number which summarizes data from a _____; also called the **point estimate**

- The value of the statistic will _____ from one sample to the next, called the _____

Parameter: a single number which summarizes data from a _____

- The value of the parameter is _____, but is typically _____
- We estimate the value of the parameter by calculating a _____

Good vs. bad sampling

GOAL: to have a sample that is _____ of the _____ on the variable(s) of interest

- If the sample is _____, then it is reasonable to believe the value of the _____ will be close to the value of the _____!
- Biased sampling method:

Types of Sampling Bias

- Selection bias:

Example of Selection Bias: Newspaper article from 1936 reported that Landon won the presidential election over Roosevelt based on a poll of 10 million voters. Roosevelt was the actual winner. What was wrong with this poll? Poll was completed using a telephone survey and not all people in 1936 had a telephone. Only a certain subset of the population owned a telephone so this subset was over-represented in the telephone survey. The results of the study, showing that Landon would win, did not represent the target population of all US voters.

- Non-response bias:

- To calculate the non-response rate:

$$\frac{\text{number of people who do not respond}}{\text{total number of people selected for the sample}} \times 100\%$$

- For non-response bias to occur must first select people to participate and then they choose not to.

Example of Non-response bias: A company randomly selects buyers to complete a review of an online purchase but some choose not to respond.

- Response bias:

Example of Response Bias: Police officer pulls you over and asks if you have been drinking. Expect people to say no, whether they have been drinking or not.

- Need to be able to predict how people will respond.

Words of caution:

- Convenience samples: gathering data for those who are easily accessible; online polls

Selection bias?

Non-response bias?

Response bias?

- Random sampling reduces _____ bias, but has no impact on _____ or _____ bias.
- Unbiased sample methods:

Simple random sample

Optional Notes: Additional Examples

Many high schools moved to partial or fully online schooling in Spring of 2020. Did students who graduated in 2020 tend to have a lower GPA during freshman year of college than the previous class of college freshmen? A nationally representative sample of 1000 college students who were freshmen in AY19-20 and 1000 college students who were freshmen in AY20-21 was taken to answer this question.

- What is the target population?
- What is the sample?

Discuss whether any type(s) of bias would be present for each of the data collection methods described below:

A radio talk show asks people to phone in their views on whether the United States should pay off its debt to the United Nations.

- Selection?
- Non-response?
- Response?

The Wall Street Journal plans to make a prediction for the US presidential election based on a survey of its readers and plans to follow-up to ensure everyone responds.

- Selection?
- Non-response?
- Response?

A police detective interested in determining the extent of drug use by high school students, randomly selects a sample of high school students and interviews each one about any illegal drug use by the student during the past year.

- Selection?
- Non-response?
- Response?

1.2.2 Concept Check

Be prepared for group discussion in the next class. One member from the table should write the answers to the following on the whiteboard.

1. What are the two types of variables?
2. Purpose of random selection:
3. Types of sampling bias:

1.3 Activity 1: Intro to Data Analysis and Sampling Bias

1.3.1 Learning outcomes

- Identify observational units, variables, and variable types in a statistical study.
- Creating a data set
- Identify biased sampling methods.

1.3.2 Terminology review

Statistics is the study of how best to collect, analyze, and draw conclusions from data. In the next few days of class, we will learn the building blocks of the semester. This week in class you will be introduced to the following terms:

- Observational units or cases
- Variables: categorical or quantitative
- Selection bias
- Response bias
- Non-response bias

For more on these concepts, read Chapter 1 and 2 in the textbook and review the Module 1 Vocabulary Review and Key Topics.

Notes on Observational Units and Variables

Review of data set

In the Fall of 2025 a data set was collected on Stat 216 students in class on day 1. The following instructions were provided to the students:

When creating a data set for use in R it is important to use single words or an underscore between words. Each outcome must be written the same way each time. Make sure to use all lowercase letters when responding to the following questions to have consistency between responses. Do not give units of measure for numerical values within the data set. For **Residency** use `in_state` or `out_state` as the two outcomes.

- Major: what is your declared major?
- Residency: do you have in-state or out-of-state residency?
- Num_Credits: how many credits are you taking this semester?
- Dominant_hand: are you left or right-handed?
- Hand_span: what is the width of your dominant hand from the tip of your thumb to the tip of your pinky with your hand spread out measured in cm?
- Grip_dominant: what is the grip strength measured in lbs for your dominant hand?
- Grip_nondominant: what is the grip strength measured in lbs for your non-dominant hand?

The following code reads in this data set and gives a snapshot of the variables collected. In a data set, the columns represent the variables collected on the observational units and there is one row of data for each observational unit.

```
data <- read.csv("data/Fall2025_DataCollection.csv")
glimpse(data)
#> Rows: 875
#> Columns: 7
#> $ Major                <chr> "Microbiology ", "Psychology ", "Business ", "~
#> $ Residency             <chr> "in state", "In state", "Out of state", "Out o~
#> $ Num_Credits           <chr> "15", "19", "17", "16", "15", "18", "16", "15"~
#> $ Dominant_hand         <chr> "right", "right", "right", "right", "right", "~
#> $ Hand_Span             <chr> "8in", "17.5 cm", "22.25", "21 cm", "19", "22"~
#> $ GripStrength_DomHand  <chr> "53.7", "54 lbs", "64", "88.3", "44.4", "86.4"~
#> $ GripStrength_NonDomHand <chr> "45.4", "32.4 lbs ", "68", "84.4", "42.2", "70~
```

1. What are the observational units or cases for the data collected?
2. How many observations are reported in the data set? This is the **sample size**.

3. Using the glimpse of the data, fill in the following table identifying the type of each variable.

- If the variable is categorical, indicate in the third column of the following table whether the variable is binary.
- If the variable is quantitative, indicate in the forth column the units of measure used.

Column	Type of Variable	Binary?	Units?
Major			
Residency			
Num Credits			
Dominant hand			
Hand Span			
Grip strength dominant hand			
Grip strength non-dominant hand			

4. Review the completed data set with your class. Remember that when creating a data set for use in R it is important to use single words or an underscore between words. Each outcome must be written the same way each time to have consistency between responses. Do not include units of measure in the data set when reporting numerical values. Write down some issues found with the created class data set.

Notes on Sampling Methods and Types of bias

Types of bias

Complete Q5 together as a class:

5. A television station is interested in predicting whether or not local voters will pass a referendum to legalize marijuana for adult use. The TV station asks its viewers to phone in and indicate whether they are in favor or opposed to the referendum. Of the 2241 viewers who phoned in, forty-five percent were opposed to legalizing marijuana.

Sample size:

Observational units sampled:

Target population:

Justify why there is selection bias in this study.

6. To determine if the proportion of out-of-state undergraduate students at Montana State University has increased in the last 10 years, a statistics instructor sent an email survey to 500 randomly selected current undergraduate students. One of the questions on the survey asked whether they had in-state or out-of-state residency. She only received 378 responses.

Sample size:

Observational units sampled:

Target population:

Justify why there is non-response bias in this study.

7. To gauge the interest of Bozeman City Voters in a new swimming pool, a local organization stood outside of the Bogert Pool in Bozeman, MT, during open hours. One of the questions they asked was, “Since the Bogert Pool is in such bad repair, don’t you agree that the city should fund a new pool?”

Sample size:

Observational units sampled:

Target population:

Justify why there is response bias in this study.

Justify why there is selection bias in this study.

1.3.3 Take-home messages

1. There are two types of variables: categorical (groups) and quantitative (numerical measures).
2. We will learn more about summarizing variables later in the semester. Categorical variables are summarized by calculating a proportion from the data and quantitative variables are summarized by finding the mean and the standard deviation (more later on this).
3. There are three types of bias to be aware of when designing a sampling method: selection bias, non-response bias, and response bias.

1.3.4 Additional notes

For more practice on types of bias, as well as identifying the observational units and variable(s) in a study, use the Module 1 Review worksheet in the Unit 1 Review Materials (pgs.108–109).

Use this space to summarize your thoughts and take additional notes on today’s activity and material covered, and to write down the names and contact information of your teammates.

1.4 Activity 2: American Indian Address

1.4.1 Learning outcomes

- Explain why a sampling method is unbiased or biased.
- Identify biased sampling methods.
- Explain the purpose of random selection and its effect on generalization.

1.4.2 Terminology review

In this activity, we will examine unbiased and biased methods of sampling. Some terms covered in this activity are:

- Random sample
- Unbiased vs biased methods of selection
- Generalization

To review these concepts, see Chapter 2 in the textbook.

1.4.3 Class Preparation

Prior to the next class, complete questions 1–3.

American Indian Address

For this activity, you will read a speech given by Jim Becenti, a member of the Navajo American Indian tribe, who spoke about the employment problems his people faced at an Office of Indian Affairs meeting in Phoenix, Arizona, on January 30, 1947 (Moquin and Van Doren 1973). His speech is below:

It is hard for us to go outside the reservation where we meet strangers. I have been off the reservation ever since I was sixteen. Today I am sorry I quit the Santa Fe [Railroad]. I worked for them in 1912–13. You are enjoying life, liberty, and happiness on the soil the American Indian had, so it is your responsibility to give us a hand, brother. Take us out of distress. I have never been to vocational school. I have very little education. I look at the white man who is a skilled laborer. When I was a young man I worked for a man in Gallup as a carpenter's helper. He treated me as his own brother. I used his tools. Then he took his tools and gave me a list of tools I should buy and I started carpentering just from what I had seen. We have no alphabetical language.

We see things with our eyes and can always remember it. I urge that we help my people to progress in skilled labor as well as common labor. The hope of my people is to change our ways and means in certain directions, so they can help you someday as taxpayers. If not, as you are going now, you will be burdened the rest of your life. The hope of my people is that you will continue to help so that we will be all over the United States and have a hand with you, and give us a brotherly hand so we will be happy as you are. Our reservation is awful small. We did not know the capacity of the range until the white man come and say "you raise too much sheep, got to go somewhere else," resulting in reduction to a skeleton where the Indians can't make a living on it. For eighty years we have been confused by the general public, and what is the condition of the Navajo today? Starvation! We are starving for education. Education is the main thing and the only thing that is going to make us able to compete with you great men here talking to us.

By eye selection

1. Circle ten words in Jim Becenti's speech which are a representative sample of the length of words in the entire text. Describe your method for selecting this sample.
2. Fill in the table below with your selected words from the previous question and the length of each word (number of letters/digits in the word):

Observation	Word	Length
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		

3. Calculate the mean (average) word length in your selected sample. Is this value a parameter or a statistic?

Notes on sampling

1.4.4 Class Activity

1. Report your mean word length from question 3 to your instructor. Your instructor will create a visualization of the distribution of results generated by your class. Draw a picture of the plot here. Include a descriptive x -axis label. Report the mean and standard deviation of the sample mean word lengths.

The plot created in question 1 is a sampling distribution of statistics. This sampling distribution plots the mean word length from many samples taken from the population of words.

2. The true mean word length of the population of all 359 words in the speech is 3.95 letters. Is this value a parameter or a statistic?

Where does the value of 3.95 fall in the plot given? Near the center of the distribution? In the tails of the distribution?

3. Would you say the sampling method used (“by-eye” selection) by the class is biased or unbiased? Justify your answer.
4. If the sampling method is biased, what type of sampling bias (selection, response, non-response) is present? What is the direction of the bias, i.e., does the method tend to overestimate or underestimate the population mean word length?

Random selection

Suppose instead of attempting to select a representative sample by eye (which did not work), each student used a random number generator to select a simple random sample of 10 words. A **simple random sample** relies on a random mechanism to choose a sample, without replacement, from the population, such that every sample of size 10 is equally likely to be chosen.

Later in the semester we will learn more about how to use R - for today we will have a short introduction using the R studio server. Download the provided R script file from canvas.

5. Login to the R studio server using your netID login. Upload the file to the server using the following instructions.
 - Click “Upload” in the “Files” tab in the bottom right window of RStudio. In the pop-up window, click “Choose File”, and navigate to the folder where the Activity R script file is saved (most likely in your downloads folder). Click “Open”; then click “Ok”.
 - Once the file is uploaded, you should see the uploaded file appear in the list of files in the bottom right window. Click on the R script file name to open the file in the Editor window (upper left window).

The following code will take a random sample of 10 words from the population of 359 words.

- In the R script file, highlight and run lines 1–7

```
sample(words$Word, 10)
```

Fill in the table with the random words selected and calculate the word length (number of letters/digits in the word):

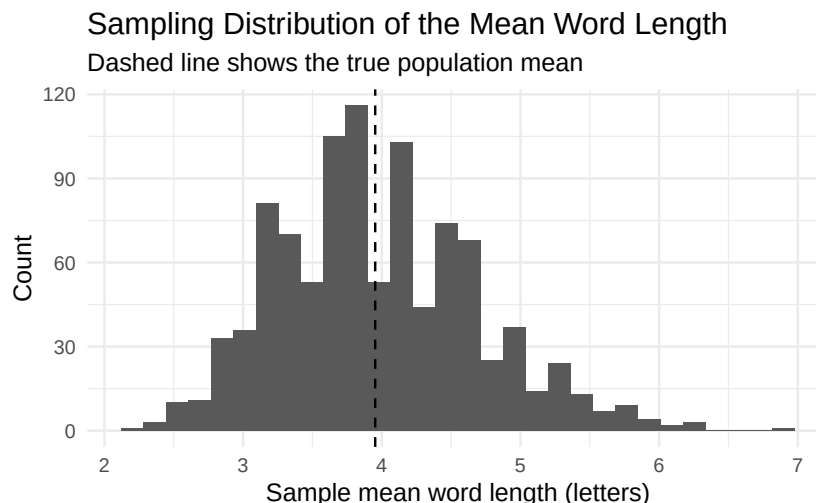
Word	Length

6. Calculate the mean word length in your selected sample above. Is this value a parameter or a statistic?

7. Report your mean word length as instructed. Your instructor will create a visualization of the distribution of results generated by your class. Draw a picture of the plot here. Include a descriptive x -axis label. Report the mean and standard deviation of the samples.

One set of randomly generated sample mean word lengths from a single class may not be large enough to visualize the distribution results.

The following is a plot of 1000 random samples of size 10 created in R. To re-create this plot run lines 11–28 in the R script file.

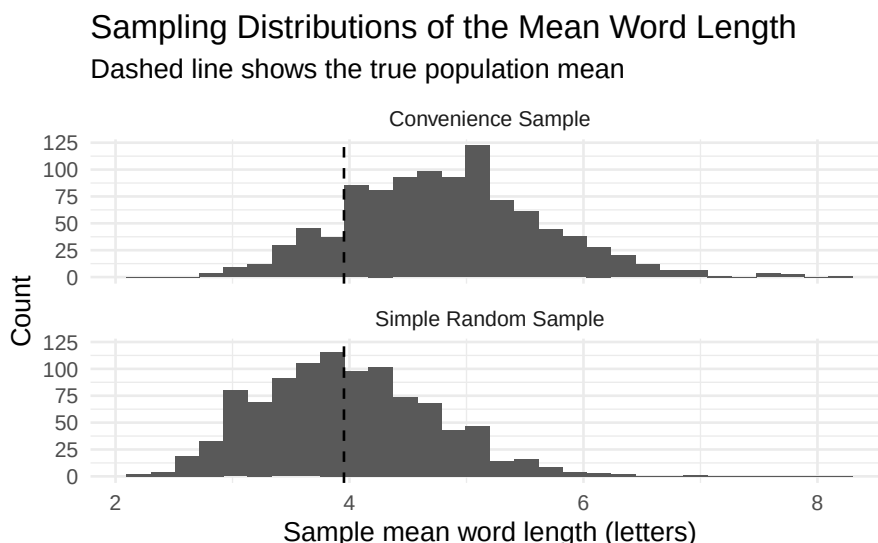


It would be useful to know what the center (mean) is for this sampling distribution. The distribution you created in the R script file may vary a little from what is reported in your coursepack. To find the mean and standard deviation from your distribution run lines 32–33.

```
mean(srs_means)
#> [1] 3.9714
sd(srs_means)
#> [1] 0.7223599
```

8. Would you say this sampling method used (random selection) is biased or unbiased? Justify your answer.

The following two plots show the sampling distribution of convenience samples of size 10 from the population of word length (top) and the simple random sample created above (bottom).



The mean and standard deviation for the convenience sample is provided below.

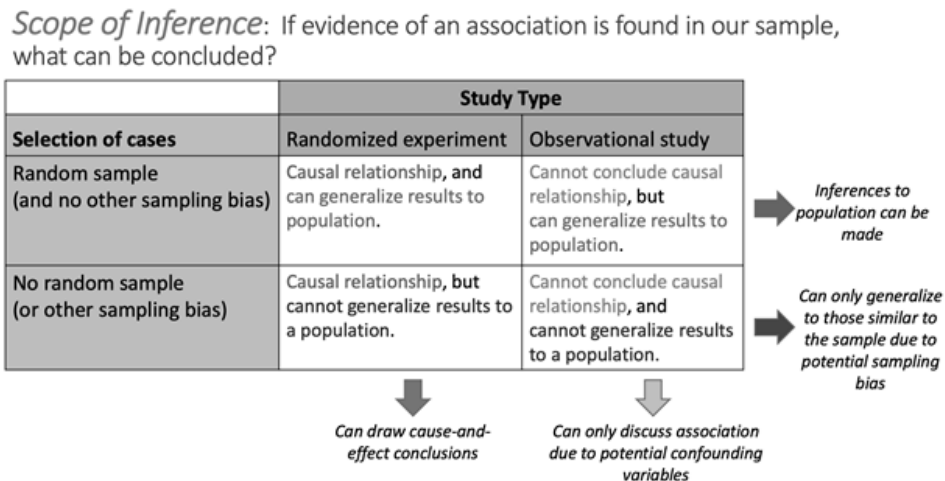
```
mean(convenience_means)
#> [1] 4.8243
sd(convenience_means)
#> [1] 0.8334471
```

9. Compare the two distributions shown above. What is similar between the two distributions? What is different?

10. Using the plots above, explain why using a random number generator to generate a sample is a “better” method than choosing 10 words “by eye” (convenience sample).

11. Is random selection an unbiased method of selection? Explain your answer. Be sure to reference the created plots.

Later in the semester we will learn about the Scope of Inference for a study. A diagram of Scope of Inference is show below:



In this activity we have focused on sampling methods. Note that the sampling method used to select the sample determines the group of observation units the results of a study apply to. This is “generalization.” We will explore causation in Module 8.

12. What is the purpose of random selection of a sample from the population?

1.4.5 Take-home messages

1. When we use a biased method of selection, we will over or underestimate the parameter, on average.
2. If the sampling method is biased, inferences made about the population based on a sample estimate will not be valid.
3. Random selection is an unbiased method of selection.
4. To determine if a sampling method is biased or unbiased, we compare the distribution of the estimates to the true value. We want our estimate to be on target or unbiased. When using unbiased methods of selection, the mean of the distribution matches or is very similar to our true parameter.
5. Random selection eliminates selection bias. However, random selection will not eliminate response or non-response bias.

1.4.6 Additional notes

For more practice on sampling methods, use the Module 1 Review worksheet in the Unit 1 Review Materials (pgs.108–109).

Use this space to summarize your thoughts and take additional notes on today's activity and material covered.

Probability

2.1 Vocabulary Review and Key Topics

2.1.1 Key topics

Module 2 introduces the concept of probability as a long-run relative frequency and demonstrates how to use hypothetical two-way tables to set up a probability problem and solve for unconditional and conditional probabilities.

2.1.2 Vocabulary

- **Probability** (of an event): the long-run proportion of times the event would occur if the random process were repeated indefinitely (under identical conditions).
- **Conditional probability** (of an event *given* another event): probability of an event calculated dependent on another event having occurred.
- **Probability notation:**
 - $P(A)$: the probability of event A .
 - * This is the probability of a single event, *unconditional* probability calculated out of the overall population.
 - $P(A^C)$: the probability of the **complement** of event A , or “ A complement”.
 - * This is the probability of the opposite of event A , or “not A ”.
 - * $P(A^C) = 1 - P(A)$
 - $P(A \text{ and } B)$: the probability of event A and B .
 - * This is the probability of an “and” event, *unconditional* probability calculated out of the overall population.
 - $P(A|B)$: the probability of event A given (conditional on) event B .
 - * This is a *conditional* probability calculated out of the total population for which event B occurred.

2.2 Video Notes: Probability

Read Chapter 23 in the course textbook. Use the following videos to complete the video notes for Module 2.

2.2.1 Course Videos

- Probability

Video: Probability (chapter 23)

Example: Two variables were collected on a random sample of people who had ever been married; whether a person had ever smoked and whether a person had ever been divorced. The data are displayed in the following table. This survey was based on a random sample in the United States in the early 1990s, so the data should be representative of the adult population who had ever been married at that time.

- Let event D be a person has gone through a divorce
- Let event S be a person smokes

	Has divorced	Has never divorced	Total
Smokes	238	247	485
Does not smoke	374	810	1184
Total	612	1057	1669

- What is the approximate probability that the person smoked?
- What is the approximate probability that the person had ever been divorced?
- Given that the person had been divorced, what is the probability that he or she smoked?
- Given that the person smoked, what is the probability that he or she had been divorced?
- Random process: one in which the outcome is _____
 - Example: Rolling a fair die
- Event: something that could occur, something we want to find the probability of; Denoted using a capital letter.
 - Example: Getting a four when rolling a fair die

- Complement: opposite of the event; Denoted using a _____
 - Example: Getting any value but a four when rolling a fair die
- The probability of an event is the _____ of times the event would occur if the random process were repeated _____; Denoted using $P(event)$
 - Example: the probability of getting a four when rolling a fair die is _____.
- Unconditional probability: calculated from the _____
 - Unconditional probabilities do **not** depend any event occurring.
 - The denominator of the probability will always be _____.
 - Two types of unconditional probabilities:
 - * The probability of a single event
 - Example from section 23.2: The probability of being late to class
 - * An “And” probability
 - Example from section 23.2: The probability of finding an open parking spot in Lot 6 **and** being late to class
- Conditional probabilities: calculated _____ on the occurrence of another event.
 - The denominator of the probability will be the total for the event _____ (or the event that is known to have occurred)
 - Example from section 23.2: The probability of being late to class, *given* that you parked in Lot 18
 - Example from section 23.2: The probability of having parked in Lot 18, *given* that you were late to class

Optional practice: For each of the four probability questions about divorce and smoking status from the beginning of these vide notes, determine if the calculated probability was a conditional or unconditional probability and write it using proper notation.

Creating a hypothetical two-way table

Steps:

- Start with a large number like 100000.
- Then use the unconditional probabilities to fill in the row or column totals.
- Now use the conditional probabilities to begin filling in the interior cells.

- Use subtraction to find the remaining interior cells.
- Add the column values together for each row to find the row totals.
- Add the row values together for each column to find the column totals.

Example: As a student at Montana State University, suppose your first class on Mondays is in Wilson Hall at 8:00am and you commute to school. You have a Bobcat parking permit. From past experience, you know that there is a 20% chance of finding an open parking spot in Lot 6 by Animal Bioscience. Otherwise, you have to park in Lot 18 by graduate housing. If you find a spot in Lot 6, you only have a 5% chance of being late to class. However, if you have to park in Lot 18, you have a 15% chance of being late to class. What is the probability that you will be late to class this Monday?

- Let event A = finding an open parking spot in Lot 6 by Animal Bioscience
- Let event C = being late to class

Start by identifying the probability notation for each value given.

- $0.20 =$
- $0.05 =$
- $0.15 =$
- Note for the table below, we will use 100,000 hypothetical Mondays (instead of 1,000 as shown in the textbook)

	C	C^c	Total
A			
A^c			
Total			100,000

For each of the following, determine if the probability is conditional or unconditional. Then write the probability in proper notation and calculate the value.

- What is the probability you are not late to class?
- What is the probability that you park in Lot 6 and you are not late to class

- Given that you were late to class, what is the probability you parked in Lot 18?

Diagnostic tests

- Sensitivity:
- Specificity:
- Prevalence:

Optional Notes: Additional Example

An airline has noticed that 30% of passengers pre-pay for checked bags at the time the ticket is purchased. The no-show rate among customers that pre-pay for checked bags is 5%, compared to 15% among customers that do not pre-pay for checked bags.

- Let event B = customer pre-pays for checked bag
- Let event N = customer no shows

Start by identifying the probability notation for each value given.

- $0.30 =$
- $0.05 =$
- $0.15 =$

	B	B^C	Total
N			
N^C			
Total			100,000

- What is the probability that a randomly selected customer who shows for the flight, pre-purchased checked bags?

2.2.2 Concept Check

Be prepared for group discussion in the next class. One member from the table should write the answers to the following on the whiteboard.

1. In the parking example, calculate and interpret the following: $P(A^C|C^C) =$.

2. What is the probability notation for 0.15 in the airline example?

2.3 Activity 3: Probability Studies

2.3.1 Learning outcomes

- Recognize and simulate probabilities as long-run frequencies.
- Construct two-way tables to evaluate conditional probabilities.

2.3.2 Terminology review

In today's activity, we will cover two-way tables and probability. Some terms covered in this activity are:

- Proportions
- Probability
- Conditional probability
- Two-way tables

To review these concepts, see Chapter 23 in the textbook.

Notes on probability

The probability of an event is the long-run proportion of times the event would occur if the random process were repeated indefinitely (under identical conditions).

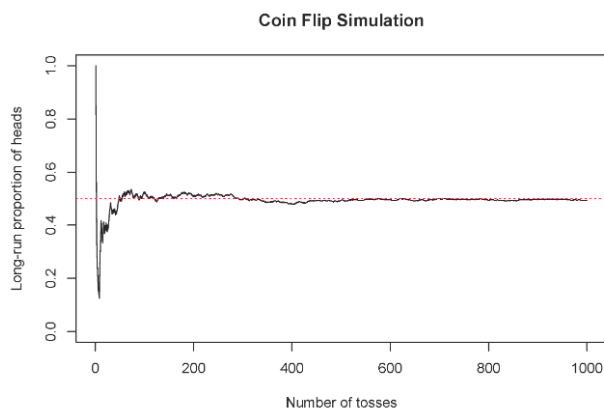
To calculate the probability of an event happening:

$$\text{probability} = \frac{\text{number of ways an event can happen}}{\text{total number of possible outcomes}}$$

For example, to calculate the probability of a coin flip landing on heads; there are only two outcomes (heads or tails) and only one possible way to land on heads.

$$P(\text{heads}) = \frac{1}{2} = 0.5$$

The figure below shows the long-run proportion of times a simulated coin flip lands on heads on the y-axis, and the number of tosses on the x-axis. Notice how the long-run proportion starts converging to 0.5 as the number of tosses increases.



In today's activity we will discuss the probability of a single event, the probability of an "and" event, and the probability of a conditional event.

Probability notation

We will use the notation $P(\text{event})$ to represent the probability of an event and use letters to represent events. The following are notations for different probabilities where we are discussing event A and event B:

- $P(A)$ represents the probability of event A
- $P(A^C)$ represents the probability of the complement of event A
 - Additionally, we can find the complement of event A by subtracting the probability of A from 1:
 $P(A^C) = 1 - P(A)$
- $P(A \text{ and } B)$ represents the probability of events A and B
- $P(A|B)$ represents the probability of event A, given event B
- $P(B|A)$ represents the probability of event B, given event A

Probability questions

For the beginning of this activity we will start with discussing the probabilities associated with drawing a card from a standard card deck. In a card deck there are:

- 52 cards
- Half are red, half are black
- Four suits: spades, hearts, diamonds, and clubs
- Each suit has 13 cards: cards 2–10, ace, jack, queen, and king
- Let A represent the event that a card is an ace
- Let B represent the event that a card is red

To find the probability of selecting an ace, first start with determining how many aces are possible (four) and how many cards will we select from (total of 52).

Find the probability of selecting a card that is not an ace. This is the complement of event A.

Find the probability of selecting a red ace.

Find the probability of selecting an ace given that the card is red.

If a card drawn is an ace, what is the probability the card drawn is red.

Calculating probabilities from a two-way table

Let's start with creating the two-way table. Your instructor will work through creating the table with the class.

1. In 2014, the website FiveThirtyEight examined the works of Bob Ross to see what trends could be found. They determined that of all the paintings he created, 95% of them contained at least one “happy tree.” Of those works with a happy tree, 43% contained at least one “almighty mountain.” Of the paintings that did not have at least one happy tree, only 10% contained at least one almighty mountain.

Let A = Bob Ross painting contains a happy tree, and B = Bob Ross painting contains an almighty mountain

Create a hypothetical two-way table to represent the situation.

	A	A^C	Total
B			
B^C			
Total			100,000

- a. What is the probability that a randomly selected Bob Ross painting contains both a “happy tree” and an “almighty mountain”? Use appropriate probability notation.
- b. What is the probability that a selected Bob Ross painting without an “almighty mountain” contains a “happy tree.” Use appropriate probability notation.
- c. What is the probability that a selected Bob Ross painting does not contain a “happy tree” given it does not contain an “almighty mountain”. Use appropriate probability notation.

2. Since the early 1980s, the rapid antigen detection test (RADT) of group A *streptococci* has been used to detect strep throat. A study of the accuracy of this test shows that the **sensitivity**, the probability of a positive RADT given the person has strep throat, is 86% in children, while the **specificity**, the probability of a negative RADT given the person does not have strep throat, is 92% in children. The **prevalence**, the probability of having group A strep, is 37% in children. (Stewart et al. 2014)

Let A = the event the child has strep throat, and B = the event the child has a positive RADT.

- a. Identify what each numerical value given in the problem represents in probability notation.

0.86 =

0.92 =

0.37 =

- b. Create a hypothetical two-way table to represent the situation.

	A	A^C	Total
B			
B^C			
Total			100,000

- c. Find $P(B)$. What does this probability represent in the context of the problem?
- d. Find the probability that a child with a positive RADT actually has strep throat. What is the notation used for this probability?
- e. What is the probability that a child does not have strep given that they have a positive RADT? What is the notation used for this probability?

2.3.3 Take home messages

1. Conditional probabilities are calculated dependent on a second variable. In probability notation, the variable following $|$ is the variable on which we are conditioning. The denominator used to calculate the probability will be the total for the variable on which we are conditioning.
2. When creating a two-way table we typically want to put the explanatory variable on the columns of the table and the response variable on the rows.
3. To fill in the two-way table, always start with the unconditional variable in the total row or column and then use the conditional probabilities to fill in the interior cells.

2.3.4 Additional notes

To understand probability you need lots of practice! Use the Module 2 Review in the Unit 1 Review Material (pgs. 110–111) for more practice problems.

Use this space to summarize your thoughts and take additional notes on today's activity and material covered.

Exploring Categorical Data: Exploratory Data Analysis and Inference using Simulation-based Methods

3.1 Vocabulary Review and Key Topics

Review the Golden Ticket posted in the resources at the end of the coursepack for a summary of a single categorical variable.

3.1.1 Key topics

Module 3 introduces the steps of the statistical investigation process. We will conduct **exploratory data analysis** (summary statistics and plots) and simulation-based **inference** (hypothesis testing and confidence intervals) in the single categorical variable (one proportion) scenario.

- Notation for a sample proportion: $\hat{p}$
- Notation for a population proportion: π
- Types of plots for a single categorical variable:
 - Frequency bar plot
 - Relative frequency bar plot

Exploratory data analysis is step 3 of the statistical investigation process. We will then use simulation-based methods **to find evidence of an effect by finding a p-value** and **estimating how large the effect is by creating a confidence interval** in the one proportion (one categorical variable) scenario. These are steps 4 and 5 from the steps of the statistical investigation process.

Steps of the statistical investigation process

As we move through the semester we will work through the six steps of the statistical investigation process.

1. Ask a research question.
2. Design a study and collect data.
3. Summarize and visualize the data.
4. Use statistical analysis methods to draw inferences from the data.
5. Communicate the results and answer the research question.
6. Revisit and look forward.

3.1.2 Vocabulary

- **Summary measure:** a numerical quantity that summarizes data. Summary measures covered in STAT 216 include: single proportion, difference in proportions, single mean, paired mean difference, difference in means, correlation, and slope of a regression line.
 - For a single categorical variable, a proportion is calculated.

- **Summary statistic (point estimate):** the value of a numerical summary measure computed from *sample* data.
 - To interpret in context include:
 - * Summary measure (in context)
 - * Value of the statistic
- **Parameter of interest:** a numerical summary measure of the entire *population* in which we are interested.
 - The value of the parameter of interest is unknown (unless we have access to the entire population).
 - To write in context:
 - * Population word (true, long-run, population)
 - * Summary measure (depends on the type of data)
 - * Context
 - Observational units
 - Variable(s)
- For a single categorical variable, the category that we are counting the proportion of is generically called a “**success**”, with categories not a success labeled “**failure**”. Thus, a sample proportion is the “proportion of successes” in the sample: the total number of successes divided by the sample size (n).

Plotting one categorical variable

- **Frequency bar plot:** plots the count (frequency) of observational units in each level of a categorical variable. R code to create a frequency bar plot:

```
object %>% # Data set piped into...
ggplot(aes(x = variable)) + # This specifies the variable
geom_bar(stat = "count") + # Tell it to make a bar plot
labs(title = "Don't forget to title your plot!",
      # Give your plot a title
      x = "x-axis label", # Label the x axis
      y = "Frequency") # Label the y axis
```

- **Relative frequency bar plot:** plots the proportion (relative frequency) of observational units in each level of a categorical variable. R code to create a relative frequency bar plot:

```
object %>% # Data set piped into...
ggplot(aes(x = variable)) + # This specifies the variable
geom_bar(aes(y = after_stat(prop), group = 1)) + # Tell it to make a bar plot with proportions
labs(title = "Don't forget to title your plot!",
      # Give your plot a title
      x = "x-axis label", # Label the x axis
      y = "Relative Frequency") # Label the y axis
```

Inference

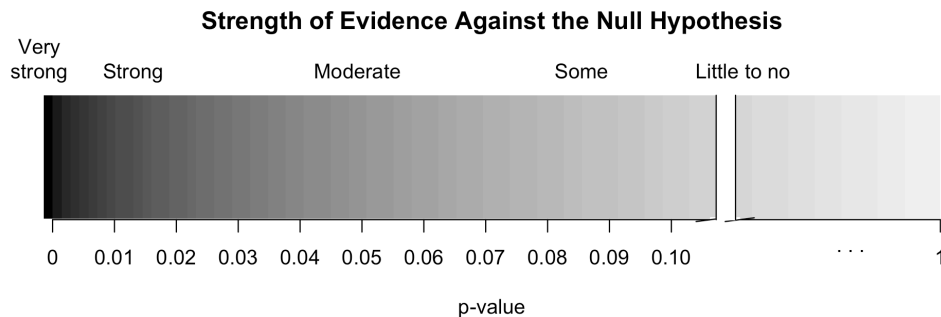
- **Sampling distribution** (of a statistic): the distribution of possible values of a statistic across repeated samples of the same size and under the same conditions.
 - We can create a *simulated* sampling distribution using simulation-based methods to simulate many samples, or we can mathematically model the sampling distribution (theory-based methods).

- **Hypothesis testing:** a formal statistical technique for evaluating two competing possibilities about a population: the null hypothesis and alternative hypothesis.
 - When we observe an effect in a sample, we would like to determine if this observed effect represents an actual effect in the population, or whether it was simply due to random chance.
 - A hypothesis test helps us answer the following question about the population: How strong is the *evidence* of an effect?
- **Null hypothesis:** typically represents a statement of “no difference”, “no effect”, or the status quo.
 - The null hypothesis is what we assume is true when calculating the p-value. Thus, we can never have evidence *for* the null hypothesis—we cannot “accept” a null hypothesis—we can only find evidence *against* the null hypothesis if the observed data is very unlikely to have occurred under the assumption that the null hypothesis is true.
- **Alternative hypothesis:** represents an alternative claim under consideration and is often represented by a range of possible values for the parameter of interest.
 - The alternative hypothesis is determined by the research question.
- **Hypotheses in notation for a single proportion:** In the hypotheses below, π_0 is the **null value**.

$$H_0 : \pi = \pi_0$$

$$H_A : \pi \left\{ \begin{array}{l} < \\ \neq \\ > \end{array} \right\} \pi_0$$

- **P-value:** the probability of the value of the observed sample statistic or a value more extreme, if the null hypothesis were true.
 - To write in context include:
 - * Statement about probability or proportion of samples
 - * Statistic (summary measure and value)
 - * Direction of the alternative
 - * Null hypothesis (in context)
- **Strength of evidence:** the p-value indicates the amount of evidence there is against the null hypothesis. The smaller the p-value the more evidence there is against the null hypothesis.



- **Conclusion** (to a hypothesis test): answers the research question. How much evidence is there in support of the alternative hypothesis?
 - To write in context include:
 - * Amount of evidence
 - * Parameter of interest
 - * Direction of the alternative hypothesis
- **Confidence interval**: an interval estimate for the parameter of interest; an interval of *plausible values* for the parameter.
 - A confidence interval helps us answer the following question about the population: How *large* is the effect?
 - To write in context include:
 - * How confident you are (e.g., 90%, 95%, 98%, 99%)
 - * Parameter of interest
 - * Calculated interval

Simulation-based inference for a single proportion

- **Conditions necessary to use simulation-based methods for inference for a single categorical variable**:
 - **Independence**: observational units must be independent of one another; the outcome of one observational unit should have no influence on the outcome of another.
- **Null distribution**: a sampling distribution of simulated sample statistics created under the assumption that the null hypothesis is true
- **Simulation-based methods to create the null distribution**: a process of using a computer program (e.g., R) to simulate many samples that we would expect based on the null hypothesis.

R code to use simulation methods for one categorical variable to find the p-value, `one_proportion_test` (from the `catstats` package), is shown below.

```
one_proportion_test(probability_success = xx, # Null hypothesis value
  sample_size = xx, # Enter sample size
  number_repetitions = 10000, # Enter number of simulations
  as_extreme_as = xx, # Observed statistic
  direction = "xx", # Specify direction of alternative hypothesis
  summary_measure = "proportion") # Reporting proportion or number of successes?
```

- **Bootstrapping**: creating a simulated sample of the same size as the original sample by sampling with replacement from the original sample.
- **Simulation-based methods to create the bootstrap distribution**: a process of using a computer program to simulate many bootstrapped samples.

R code to use simulation methods for one categorical variable to find a confidence interval, `one_proportion_bootstrap_CI` (from the `catstats` package), is shown below.

```
one_proportion_bootstrap_CI(sample_size = xx, # Sample size
  number_successes = xx, # Observed number of successes
  number_repetitions = 10000, # Number of bootstrap samples to use
  confidence_level = 0.95) # Confidence level as a decimal
```


- **Percentile method:** process to find the confidence interval from the bootstrap distribution.
 - A 90% confidence interval will be found between the 5th and 95th percentiles of the bootstrap distribution.
 - A 95% confidence interval will be found between the 2.5th and 97.5th percentiles of the bootstrap distribution.
 - A 99% confidence interval will be found between the 0.5th and 99.5th percentiles of the bootstrap distribution.

3.2 Video Notes: Analysis of One Categorical Variable

Read Chapter 3, 4, 9, 10 and Sections 14.1 and 14.2 in the course textbook. Use the following videos to complete the video notes for Module 3.

3.2.1 Course Videos

- One_Categorical_Variable_EDA
- Hypothesis_Testing
- Simulation_Tests_One_Categorical_Variable
- Confidence_Intervals
- Bootstrap_Intervals_One_Categorical_Variable

Video: One Categorical Variable EDA (sections 4.1 and 4.2 - summarizing and displaying one categorical variable)

- A _____ is calculated on data from a sample
- The _____ is calculated on data from a population; it is typically unknown and therefore we will define it *in words*
- The parameter should include:
 - Population word (true, long-run, population)
 - Summary measure (depends on the type of data)
 - Context
 - * Observational units
 - * Variable(s)

Categorical data can be numerically summarized by calculating a _____ from the data set.

Notation used for the population proportion:

Notation used for the sample proportion:

Categorical data can be summarized in a _____ table, which reports counts or a _____ table, which reports proportions.

Types of plots for a single categorical variable:

Example from section 4.1 using the **email** data set:

- Identify the observational units.
- What type of variable is **number**?

- Is Table 4.3 a frequency or relative frequency table?
- What proportion of emails had no number in them? Is this value a statistic or a parameter? Write it using proper notation.
- Compare the two plots (frequency bar plot and relative frequency bar plot) of the variable **number** in Figure 4.1 in section 4.2 of the text. Note that the x-axis is the _____ in the two plots. However, the _____ differs. The scale for the frequency bar plot goes from _____ and the scale for the relative frequency bar plot is from _____.

Optional Notes: Additional Example

Gallatin Valley is the fastest growing county in Montana. You'll often hear Bozeman residents complaining about the 'out-of-staters' moving in. A local real estate agent recorded data on a random sample of 100 home sales over the last year at her company and noted where the buyers were moving from as well as the age of the person or average age of a couple buying a home. The variable age was binned into two categories, "Under30" and "Over30." Additionally, the variable, state the buyers were moving from, was created as a binary variable, "Out" for a location out of state and "In" for a location in state.

The following code reads in the data set, `moving_to_mt` and names the object `moving`.

```
moving <- read.csv("data/moving_to_mt.csv")
```

The R function `glimpse` was used to give the following output.

```
glimpse(moving)
```

```
#> Rows: 100
#> Columns: 4
#> $ From      <chr> "CA", "CA", "CA", "CA", "CA", "CA", "CA", "CA", "CA", "CA", ~
#> $ Age_Group <chr> "Under30", "Under30", "Under30", "Under30", "Under30", "Unde~
#> $ Age       <int> 25, 26, 27, 27, 29, 29, 35, 37, 49, 63, 65, 77, 22, 24, 24, ~
#> $ InOut     <chr> "Out", "Out", "Out", "Out", "Out", "Out", "Out", "Out", "Out", "Out~
```

- What are the observational units in this study?
- What type of variable is `Age`?

- What type of variable is `Age_Group`?

To further analyze the categorical variable, `From`, we can create either a frequency table:

```
#>   From  n
#> 1   CA 12
#> 2   CO  8
#> 3   MT 61
#> 4   WA 19
```

Or a relative frequency table:

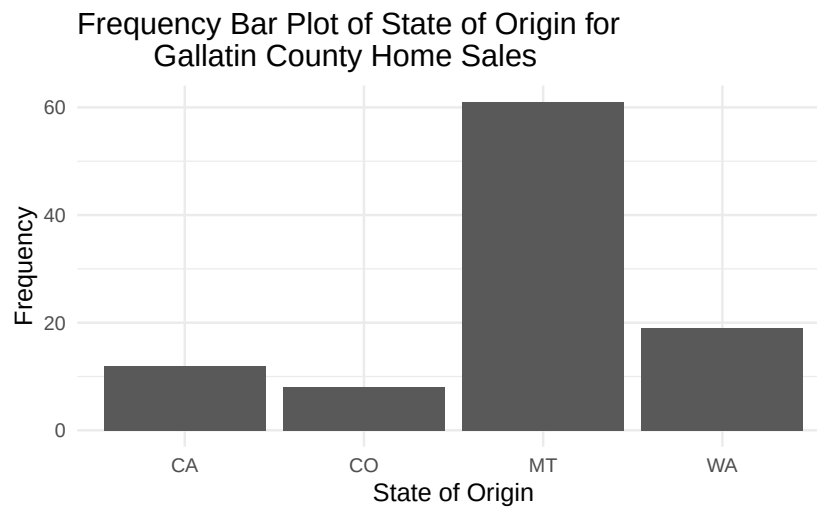
```
#>   From  n freq
#> 1   CA 12 0.12
#> 2   CO  8 0.08
#> 3   MT 61 0.61
#> 4   WA 19 0.19
```

- How many home sales have buyers from WA?
- What proportion of sampled home sales have buyers from WA?
- What notation is used for the proportion of home sale buyers that are from WA?

The following code in R will create a frequency bar plot of the variable, `From`.

moving `%>%`

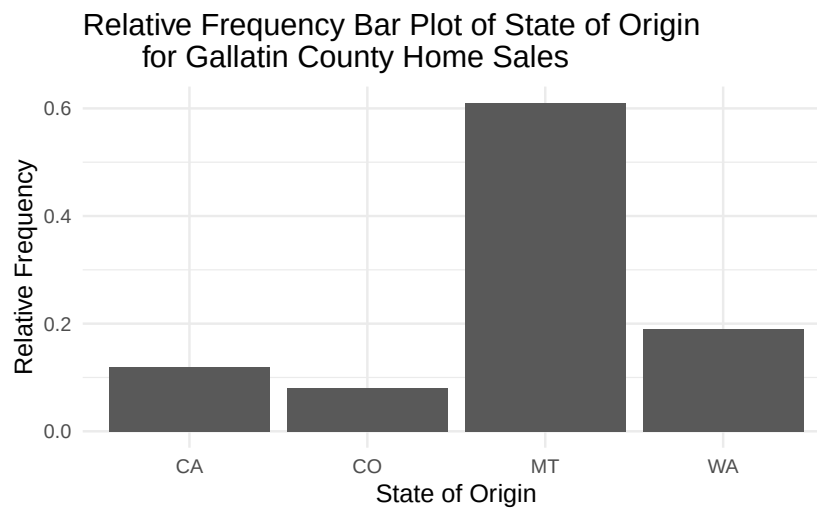
```
ggplot(aes(x = From)) + #Enter the variable to plot
  geom_bar(stat = "count") +
  labs(title = "Frequency Bar Plot of State of Origin for
    Gallatin County Home Sales",
    #Title your plot (type of plot, observational units, variable)
    y = "Frequency", #y-axis label
    x = "State of Origin") #x-axis label
```



- What can we see from this plot?

Additionally, we can create a relative frequency bar plot.

```
moving %>%
  ggplot(aes(x = From)) + #Enter the variable to plot
  geom_bar(aes(y = after_stat(prop), group = 1)) +
  labs(title = "Relative Frequency Bar Plot of State of Origin
    for Gallatin County Home Sales",
    #Title your plot
    y = "Relative Frequency", #y-axis label
    x = "State of Origin") #x-axis label
```



Video: Hypothesis Testing (chapter 9)

Purpose of a hypothesis test:

- Use data collected on a sample to give information about the population.
- Determines _____ of _____ of an effect

General steps of a hypothesis test

1. Write a research question and hypotheses.
2. Collect data and calculate a summary statistic.
3. Model a sampling distribution which assumes the null hypothesis is true.
4. Calculate a p-value.
5. Draw conclusions based on a p-value.

Hypotheses

- Always hypothesize about the _____, so will always be written in terms of _____.
- Both null and alternative hypotheses will compare the parameter to the same value, only the *sign* will change.

Null hypothesis

- Skeptical perspective, no difference, no effect, random chance
- What the researcher hopes is _____.
- Will always use the _____ sign.

Notation:

Alternative hypothesis

- New perspective, a chance, a difference, an effect
- What the researcher hopes is _____.
- The sign of the alternative hypothesis reflects the _____.

Notation (*write all three options*):

Simulation vs. Theory-based Methods

Simulation-based method

Creation of the null distribution

- Simulate many samples assuming

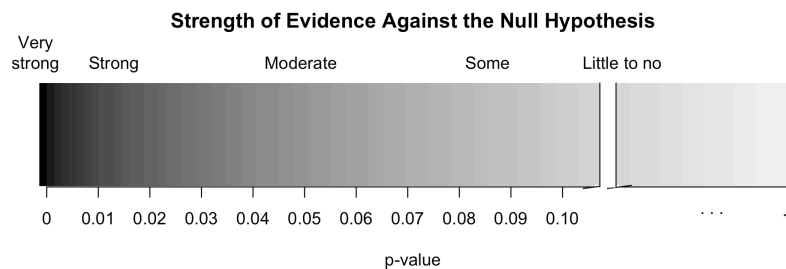
- Find the proportion of _____ at least as extreme as the observed sample _____
- The null distribution estimates the sample to sample variability expected in the population

Theory-based method

- Use a mathematical model to determine a distribution under the null hypothesis
- Compare the observed sample statistic to the model to calculate a probability
- *Theory-based methods will be discussed in the next module*

P-value

- What does the p-value measure?
 - Probability of observing the sample _____ or more _____ assuming the _____ hypothesis is _____.
- How much evidence does the p-value provide against the null hypothesis?



- The _____ the p-value, the _____ the evidence against the null hypothesis.

- Write a conclusion based on the p-value.
 - Answers the _____ question.
 - Amount of _____ in support of the _____ hypothesis.
- Decision: can we reject or fail to reject the null hypothesis?
 - Significance level: cut-off of “small” vs “large” p-value
 - $p\text{-value} \leq \alpha$
 - Strong enough evidence against the null hypothesis
 - Decision:

- Results are _____ significant.
- $p\text{-value} > \alpha$
- Not enough evidence against the null hypothesis
- Decision:
- Results are not _____ significant.

Hypothesis Testing/Justice System

- Null hypothesis: The defendant is _____
- Alternative hypothesis: The defendant is _____
- Prosecution collects evidence (data) and presents that evidence to a jury
- Two possible outcomes:
 - Strong evidence against _____ -> jury finds the defendant _____
 - Not enough evidence against _____ -> jury finds the defendant _____

Example: Martian Alphabet

Review the Martian Alphabet example from sections 9.1 and 9.3.2

- Write the null hypothesis in words and in notation:

In words: H_0 :

In notation: $H_0 : \pi = \underline{\hspace{2cm}}$, where π represents the _____ proportion of humans that would _____.

- Write the alternative hypothesis in words and in notation:

In words: H_A :

In notation: $H_A : \pi \underline{\hspace{2cm}}$ (a _____ sign represents the research question - that humans have a _____ for choosing Bumba on the left)

- What is the value of the statistic? Write it using proper notation.

- Review Figure 9.2. Why is the plot centered around 0.5? What does each “dot” on the plot represent?
- Does the statistic seem likely or unlikely to occur if humans are picking a shape randomly? What about Figure 9.2 helps us to answer this question?
- The p-value of the test is _____. This p-value provides _____ evidence against the _____ hypothesis and for the _____ hypothesis. The results from this study _____ be considered statistically significant.

Video: Simulation Tests for One Categorical Variable (section 14.1)

Review the Medical Consultant example from section 14.1.

Identify the observational units.

Identify the variable collected.

Is the variable categorical or quantitative? If categorical, define a “success”; if quantitative, state the units of measure.

Define the parameter in words and write it using proper notation

Write the null and alternative hypotheses in words and in proper notation:

In words:

H_0 :

H_A :

In notation:

$$H_0 :$$

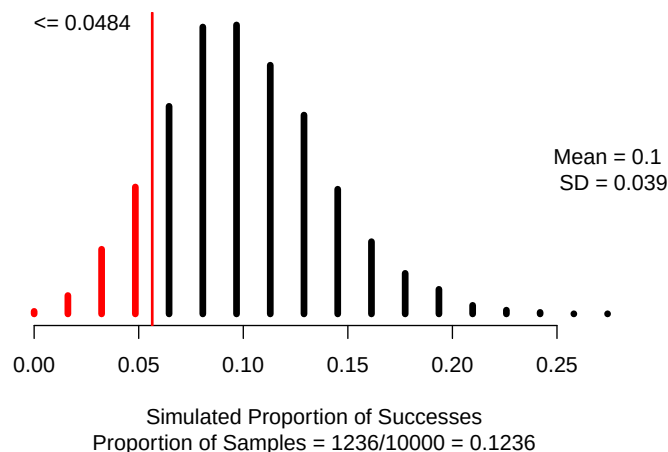
$$H_A :$$

Calculate the value of the statistic and write it using proper notation:

Simulation-based method

- Simulate many samples assuming $H_0 : \pi = \pi_0$
 - Create a spinner with that represents the null value
 - Spin the spinner n times
 - Calculate and plot the simulated sample proportion from each simulation
 - Repeat 10000 times (simulations) to create the null distribution
 - Find the proportion of simulations at least as extreme as $\hat{p}$

```
set.seed(216)
one_proportion_test(probability_success = 0.10, # Null hypothesis value
  sample_size = 62, # Enter sample size
  number_repetitions = 10000, # Enter number of simulations
  as_extreme_as = 0.0484, # Observed statistic
  direction = "less", # Specify direction of alternative hypothesis
  summary_measure = "proportion") # Reporting proportion or number of successes?
```



Explain why the null distribution is centered at the value of approximately 0.10:

Where is the red line plotted on the null distribution? Why is that value important?

Why is the area left of the red line shaded on the null distribution?

Interpretation of the p-value:

- Statement about probability or proportion of samples
- Statistic (summary measure and value) and Direction of the alternative
- Null hypothesis (population reference, summary measure, equal to null value)
- Context of the problem (observational units, variable(s))

Two example p-value interpretations are provided below. Identify each of the components within the two examples:

- Example 1: There is a 12.36% probability of observing three (3) or fewer complications out of sixty-two (63) liver donor surgeries in the US, if the true proportion of complications is 0.10.
- Example 2: If 10% of all liver donor surgeries in the US result in a complication, 1236/10000 samples would simulate a proportion of complications of 0.048 or less.

Conclusion:

- Amount of evidence
- For the alternative hypothesis (population reference, summary measure, direction, null value)
- Context (observational units, variable(s))

An example conclusion is provided below. Identify each of the components within the example:

- There is little to no that the proportion of all liver donor surgeries in the US that result in a complication is less than 0.10.

Generalization:

- Can the results of the study be generalized to the target population?

Video: Confidence intervals (chapter 10)

statistic $\pm$ margin of error

Vocabulary:

- Point estimate:

- Margin of error:

Purpose of a confidence interval

- To give an _____ for the parameter of interest
- Determines how _____ an effect is

Sampling distribution

- Ideally, we would take many samples of the same _____ from the same population to create a sampling distribution
- But only have 1 sample, so we will _____ with _____ from the one sample.
- Need to estimate the sampling distribution to see the _____ in the sample

Simulation-based methods

Bootstrap distribution:

- Write the response variable values on cards
- Sample with replacement n times (bootstrapping)
- Calculate and plot the simulated difference in sample means from each simulation
- Repeat 10000 times (simulations) to create the bootstrap distribution
- Find the cut-offs for the middle X% (confidence level) in a bootstrap distribution.

What is bootstrapping?

- Assume the “population” is many, many copies of the original sample.
- Randomly sample with replacement from the original sample n times.

Example: Medical Consultant

Review the Medical Consultant example in section 10.1.

- Is 0.048 the statistic or the parameter? How do you know?

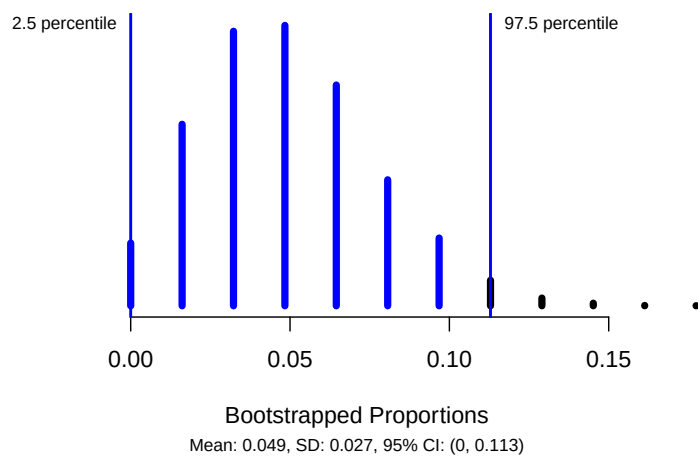
- Review Figure 10.6 in section 10.1.2.
 - Why is the plot centered near 0.048? What does each “dot” on the plot represent?
 - Why are the 2.5th and 97.5th percentiles of the bootstrap distribution marked? What are the values of these two percentiles?
- Interpret the confidence interval: We are _____% _____ that, in the _____, the true probability (or proportion) of complications is between _____ and _____.

Video: Bootstrap Confidence Intervals for One Categorical Variable (section 14.2)

Review the Medical Consultant study from section 14.2.

Bootstrap distribution:

```
set.seed(216)
one_proportion_bootstrap_CI(sample_size = 62, # Sample size
                             number_successes = 3, # Observed number of successes
                             number_repetitions = 10000, # Number of bootstrap samples to use
                             confidence_level = 0.95) # Confidence level as a decimal
```



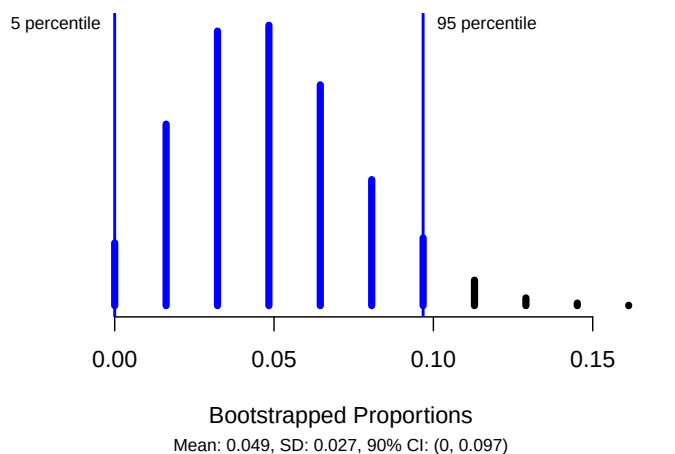
Confidence interval interpretation:

- How confident you are (e.g., 90%, 95%, 98%, 99%)
- Parameter of interest
- Calculated interval

How does changing the confidence level impact the width of the confidence interval?

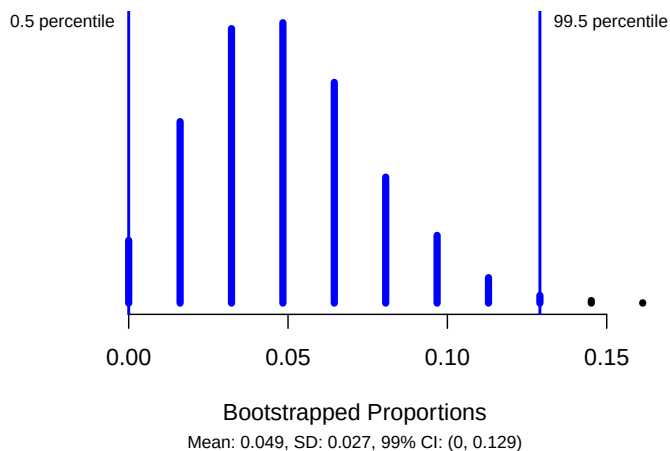
90% Confidence Interval:

```
set.seed(216)
one_proportion_bootstrap_CI(sample_size = 62, # Sample size
                             number_successes = 3, # Observed number of successes
                             number_repetitions = 10000, # Number of bootstrap samples to use
                             confidence_level = 0.90) # Confidence level as a decimal
```



99% Confidence Interval:

```
set.seed(216)
one_proportion_bootstrap_CI(sample_size = 62, # Sample size
                             number_successes = 3, # Observed number of successes
                             number_repetitions = 10000, # Number of bootstrap samples to use
                             confidence_level = 0.99) # Confidence level as a decimal
```



3.2.2 Concept Check

Be prepared for group discussion in the next class. One member from the table should write the answers to the following on the whiteboard.

1. What is the summary measure calculated from a single categorical variable?
2. Write the alternative hypothesis for the Martian Alphabet study in notation? How was the direction of the alternative hypothesis determined?
3. Do the results of the confidence interval for the Medical Consultant study *match* the results based on the p-value?

3.3 Activity 4: Helper-Hinderer Part 1 — Simulation-based Hypothesis Test

3.3.1 Learning outcomes

- Identify the two possible explanations (one assuming the null hypothesis and one assuming the alternative hypothesis) for a relationship seen in sample data.
- Given a research question involving a single categorical variable, construct the null and alternative hypotheses in words and using appropriate statistical symbols.
- Describe and perform a simulation-based hypothesis test for a single proportion.

3.3.2 Terminology review

In today's activity, we will work through a simulation-based hypothesis testing for a single categorical variable. Some terms covered in this activity are:

- Parameter of interest
- Null hypothesis
- Alternative hypothesis
- Simulation

To review these concepts, see Chapters 9 & 14 in your textbook.

3.3.3 Steps of the statistical investigation process

We will work through a five-step process to complete a hypothesis test for a single proportion.

- **Ask a research question** that can be addressed by collecting data. What are the researchers trying to show?
- **Design a study and collect data.** This step involves selecting the people or objects to be studied and how to gather relevant data on them.
- **Summarize and visualize the data.** Calculate summary statistics and create graphical plots that best represent the research question.
- **Use statistical analysis methods to draw inferences from the data.** Choose a statistical inference method appropriate for the data and identify the p-value and/or confidence interval after checking assumptions. In this study, we will focus on using randomization to generate a simulated p-value.
- **Communicate the results and answer the research question.** Using the p-value and confidence interval from the analysis, determine whether the data provide statistical evidence against the null hypothesis. Write a conclusion that addresses the research question.

Notes on one categorical variable

3.3.4 Helper-Hinderer

A study by Hamblin, Wynn, and Bloom reported in Nature (Hamblin et al. 2007) was intended to check young kids' feelings about helpful and non-helpful behavior. Non-verbal infants ages 6 to 10 months were shown short videos with different shapes either helping or hindering the climber. As a class we will watch this short video to see how the experiment was run: <https://youtu.be/anCaGBsBOxM>. Researchers were hoping to assess: Are non-verbal infants more likely to choose the helper toy? In the study, of the 16 infants age 6 to 10 months, 14 chose the *helper* toy and 2 chose the *hinderer* toy.

- Observational units:
- Variable:
 - Success:

Ask a research question

- Research question:

Design a study and collect data

Before using statistical inference methods, we must check that the cases are independent. The sample observations are independent if the outcome of one observation does not influence the outcome of another. One way this condition is met is if data come from a simple random sample of the target population.

1. Are the cases independent? Justify your answer.

R Instructions

For almost all activities and labs it will be necessary to upload the provided R script file from Canvas for that day. Your instructor will highlight a few steps in uploading files to and using RStudio.

The following are the steps to upload the necessary R script file for this activity:

- Download the Activity R script file from Canvas.

- Click “Upload” in the “Files” tab in the bottom right window of RStudio. In the pop-up window, click “Choose File”, and navigate to the folder where the Activity R script file is saved (most likely in your downloads folder). Click “Open”; then click “Ok”.
- You should see the uploaded file appear in the list of files in the bottom right window. Click on the file name to open the file in the Editor window (upper left window).

Notice that the first three lines of code contain a prompt called **library**. Packages needed to run functions in R are stored in directories called libraries. When using the MSU RStudio server, all the packages needed for the class are already installed. We simply must tell R which packages we need for each R script file. We use the prompt **library** to load each **package** (or library) needed for each activity. Note, these **library** lines MUST be run each time you open a R script file in order for the functions in R to work.

- Highlight and run lines 1–3 to load the packages needed for this activity. Notice the use of the **#** symbol in the R script file. This symbol is not part of the R code. It is used by these authors to add comments to the R code and explain what each call is telling the program to do.

R will ignore everything after a **#** symbol when executing the code. Refer to the instructions following the **#** symbol to understand what you need to enter in the code.

```
library(tidyverse)
library(ggplot2)
library(catstats)
```

Throughout activities, we will often include the R code you would use in order to produce output or plots. These “code chunks” appear in gray. In the code chunk below, we demonstrate how to read the data set into R using the **read.csv()** function. The line of code shown below (line 7 in the R script file) reads in the data set and names the data set **infants**.

Summarize and visualize the data

The following code reads in the data set and gives the number of infants in each level of the variable, whether the infant chose the helper or the hinderer.

- Highlight and run lines 7 and 8 to check that you get the same counts as shown below

```
# Read in data set
infants <- read.csv("https://math.montana.edu/courses/s216/data/infantchoice.csv")
infants %>% count(choice) # Count number in each choice category
```

```
#>      choice    n
#> 1    helper  14
#> 2 hinderer   2
```

The following formula is used to calculate the proportion of successes in the sample.

$$\hat{p} = \frac{\text{number of successes}}{\text{total number of observational units}}$$

2. Using the R output and the formula given, calculate the summary statistic (sample proportion) to represent the research question. Recall that **choosing the helper toy** is a considered a success. Use appropriate notation.

To visually display this data we can use either a frequency bar plot or a relative frequency bar plot.

- Enter the variable name **choice** for **variable** in the R code to create the frequency bar plot.
- Note the name of the title is given in line 16 and includes the **type of plot**, **observational units**, and **variable name**.
- Highlight and run lines 13–19 to create the plot

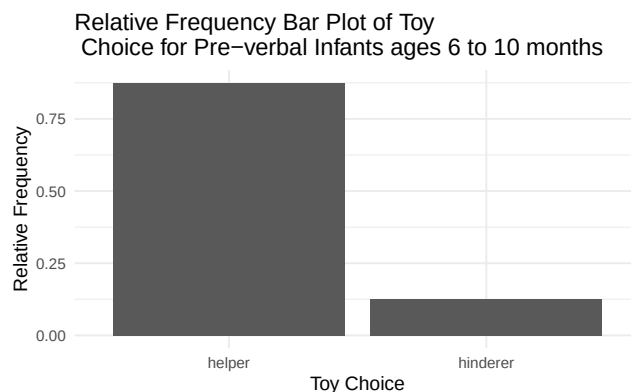
```
infants %>% # Data set piped into...
  ggplot(aes(x = variable)) + # This specifies the variable
  geom_bar(stat = "count") + # Tell it to make a bar plot
  labs(title = "Frequency Bar Plot of Toy Choice for Pre-verbal Infants ages 6 to 10 months",
        # Give your plot a title
        x = "Toy Choice", # Label the x axis
        y = "Frequency") # Label the y axis
```

3. Sketch the frequency bar plot created below.

We could also choose to display the data as a proportion in a **relative frequency** bar plot. To find the relative frequency, the count in each level of **choice** is divided by the sample size. This calculation is the sample proportion for each level of **choice**. Notice that in the following code we told R to create a bar plot with proportions.

- In the R script file, highlight and run lines 23–29 to create the relative frequency bar plot.

```
infants %>% # Data set piped into...
  ggplot(aes(x = choice)) + # This specifies the variable
  geom_bar(aes(y = after_stat(prop), group = 1)) + # Tell it to make a bar plot with proportions
  labs(title = "Relative Frequency Bar Plot of Toy Choice for Pre-verbal Infants ages 6 to 10 months",
        # Give your plot a title
        x = "Toy Choice", # Label the x axis
        y = "Relative Frequency") # Label the y axis
```



4. Which features in the relative frequency bar plot are the same as the frequency bar plot? Which are different?

We cannot assess whether infants are more likely to choose the helper toy based on the statistic and plot alone. The next step is to analyze the data by using a hypothesis test to discover if there is evidence against the null hypothesis.

Use statistical analysis methods to draw inferences from the data

Notes on hypotheses tests

When performing a hypothesis test, we must first identify the null hypothesis. The null hypothesis is written about the parameter of interest, or the value that summarizes the variable in the population.

The parameter of interest is a statement about what we want to find about the population. The following must be included when writing the parameter of interest.

- Population word (true, long-run, population)
- Summary measure (depends on the type of data)
- Context
 - Observational units
 - Variable(s)

Parameter of interest:

If the children are just randomly choosing the toy, we would expect half (0.5) of the infants to choose the helper toy. This is the null value for our study.

Null Hypothesis (in words):

The notation used for a population proportion (or probability, or true proportion) is π . Since this summarizes a population, it is a parameter. When writing the **null hypothesis** in notation, we set the parameter equal to the null value, $H_0 : \pi = \pi_0$.

Null Hypothesis (in notation):

The **alternative hypothesis** is the claim to be tested and the direction of the claim (less than, greater than, or not equal to) is based on the research question.

- Based on the research question from question 1, are we testing that the parameter is greater than 0.5, less than 0.5 or different than 0.5?

Alternative hypothesis (in words):

Alternative hypothesis (in notation):

Remember that when utilizing a hypothesis test, we are evaluating two competing possibilities. For this study the **two possibilities** are either...

- The true proportion of infants who choose the helper is 0.5 and our results just occurred by random chance; or,
- The true proportion of infants who choose the helper is greater than 0.5 and our results reflect this.

Notice that these two competing possibilities represent the null and alternative hypotheses.

We will now simulate one sample of a **null distribution** of sample proportions. The null distribution is created under the assumption the null hypothesis is true. In this case, we assume the true proportion of infants who choose the helper is 0.5, so we will create 10000 (or more) different simulations of 16 infants under this assumption.

Let's think about how to use a coin to create one simulation of 16 infants under the assumption the null hypothesis is true. Let heads equal infant chose the helper toy and tails equal infant chose the hinderer toy.

5. How many times would you flip a coin to simulate the sample of infants?
6. Flip a coin 16 times recording the number of times the coin lands on heads. This represents one simulated sample of 16 infants randomly choosing the toy. Calculate the proportion of coin flips that resulted in heads.
7. Is the value from question 6 closer to 0.5, the null value, or closer to the sample proportion, 0.875?

Report the number of coin flips you got as indicated by your instructor.

8. Sketch the graph created by your instructor of each student's proportion of heads out of 16 coin flips.

9. Circle the observed statistic (value from question 2) on the distribution shown above. Where does this statistic fall in this distribution: Is it near the center of the distribution (near 0.5) or in one of the tails of the distribution?
10. Is the observed statistic likely to happen or unlikely to happen if the true proportion of infants who choose the helper is 0.5? Explain your answer using the plot.

In the next class, we will continue to assess the strength of evidence against the null hypothesis by using a computer to simulate 10000 samples when we assume the null hypothesis is true.

3.3.5 Take-home messages

1. Two types of plots are used for plotting categorical variables: frequency bar plots, relative frequency bar plots.
2. In a hypothesis test we have two competing hypotheses, the null hypothesis and the alternative hypothesis. The null hypothesis represents either a skeptical perspective or a perspective of no difference or no effect. The alternative hypothesis represents a new perspective such as the possibility that there has been a change or that there is a treatment effect in an experiment.
3. In a simulation-based test, we create a distribution of possible simulated statistics for our sample if the null hypothesis is true. Then we see if the calculated observed statistic from the data is likely or unlikely to occur when compared to the null distribution.
4. To create one simulated sample on the null distribution for a sample proportion, spin a spinner with probability equal to π_0 (the null value), n times or draw with replacement n times from a deck of cards created to reflect π_0 as the probability of success. Calculate and plot the proportion of successes from the simulated sample.

3.3.6 Additional notes

Activities 4–6 cover the material in Module 3. Use this space to summarize your thoughts and take additional notes on today's activity and material covered.

3.4 Activity 5: Helper-Hinderer (continued)

3.4.1 Learning outcomes

- Describe and perform a simulation-based hypothesis test for a single proportion.
- Interpret and evaluate a p-value for a simulation-based hypothesis test for a single proportion.
- Explore what a p-value represents

3.4.2 Steps of the statistical investigation process

In today's activity we will continue with steps 4 and 5 in the statistical investigation process. We will continue to assess the Helper-Hinderer study from last class.

- **Ask a research question** that can be addressed by collecting data. What are the researchers trying to show?
- **Design a study and collect data.** This step involves selecting the people or objects to be studied and how to gather relevant data on them.
- **Summarize and visualize the data.** Calculate summary statistics and create graphical plots that best represent the research question.
- **Use statistical analysis methods to draw inferences from the data.** Choose a statistical inference method appropriate for the data and identify the p-value and/or confidence interval after checking assumptions. In this study, we will focus on using randomization to generate a simulated p-value.
- **Communicate the results and answer the research question.** Using the p-value and confidence interval from the analysis, determine whether the data provide statistical evidence against the null hypothesis. Write a conclusion that addresses the research question.

3.4.3 Helper-Hinderer

In class today, we will revisit the study on infants as described below.

A study by Hamblin, Wynn, and Bloom reported in Nature (Hamblin et al. 2007) was intended to check young kids' feelings about helpful and non-helpful behavior. Non-verbal infants ages 6 to 10 months were shown short videos with different shapes either helping or hindering the climber. As a class we will watch this short video to see how the experiment was run: <https://youtu.be/anCaGBsBOxM>. Researchers were hoping to assess: Are non-verbal infants more likely to choose the helper toy? In the study, of the 16 infants age 6 to 10 months, 14 chose the *helper* toy and 2 chose the *hinderer* toy.

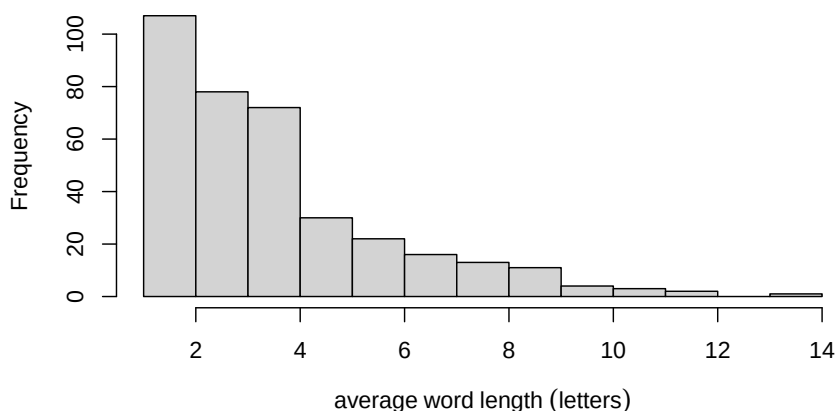
Sampling distribution of the sample proportion.

The sampling distribution is a distribution of a statistic from all possible samples of size n from the population. The mean of the sampling distribution should be close to the parameter value and the standard deviation of the sampling distribution will estimate the variability of statistics from sample to sample.

To create a sampling distribution of the statistic, we would take infinite samples of size n from the population and plot the statistic from each sample. To illustrate this idea we will revisit the Becenti address looking at the mean word length.

```
#>   min Q1 median Q3 max      mean      sd    n missing
#> 1   1   2      3   5  14 3.952646 2.300228 359      0
```

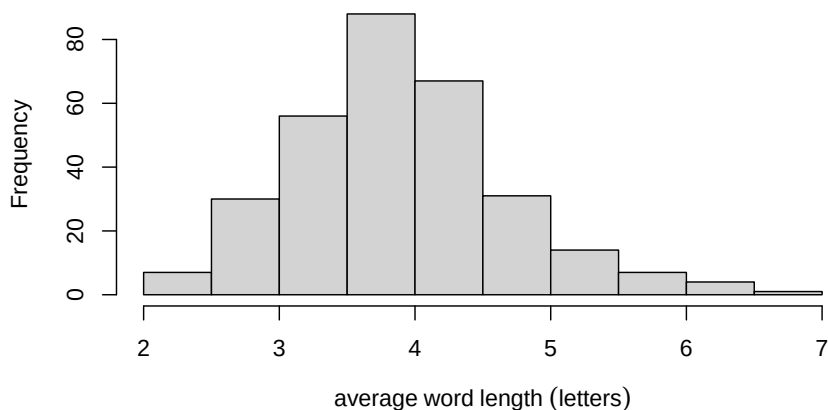
Histogram of the Population Distribution



Note that the mean of this small population ($N = 359$) is 3.9526. Typically, we don't have such small population sizes so we need a way to simulate the sampling distribution of the statistic to estimate the sample to sample variability. Repeat samples of size 10 are taken from the population of words and the sample mean of each sample is plotted below.

```
#>   min  Q1 median  Q3 max    mean      sd  n missing
#> 1  2.1  3.3    3.9  4.4  6.7  3.934557 0.7954711 305      0
```

Histogram of the Sampling Distribution of the Average



The mean of this simulated distribution is 3.934557, which is close to the true mean from the population. The standard deviation is 0.7954711 - this measures how far each statistic is from the true mean word length, on average. Later in the semester, we will learn how this variability is affected by the sample size.

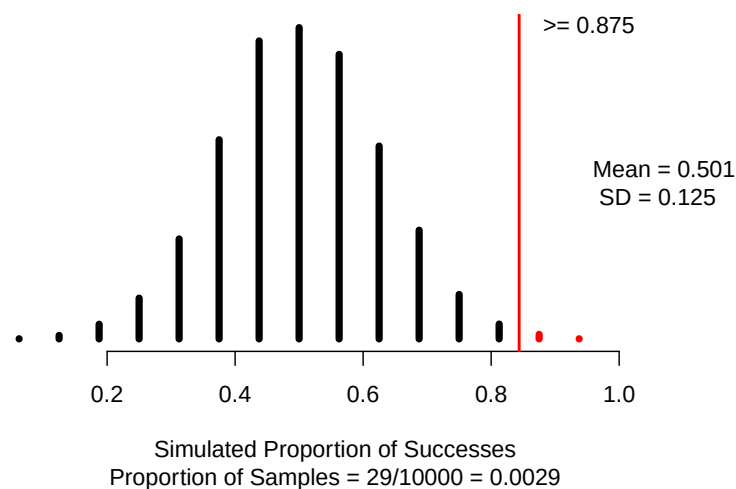
Today, we will use the computer to simulate the sampling distribution of the sample proportion by creating a null distribution of 10000 different samples of 16 infants, plotting the proportion who will choose the helper in each sample, based on the assumption that the true proportion of infants who choose the helper is 0.5 (or that the null hypothesis is true).

To use the computer simulation, we will need to enter the

- assumed “probability of success” (π_0),
 - “sample size” (the number of observational units or cases in the sample),
 - “number of repetitions” (the number of samples to be generated - typically we use 10000),
 - “as extreme as” (the observed statistic), and
 - the “direction” (matches the direction of the alternative hypothesis).
1. What values should be entered for each of the following into the one proportion test to create 10000 simulations?
 - Probability of success (null value):
 - Sample size (n):
 - Number of repetitions (typically use 10000 simulations):
 - As extreme as (value of statistic):
 - Direction ("greater", "less", or "two-sided"):

The following shows one possible distribution of simulated sample proportions assuming the null hypothesis, true proportion of non-verbal infants aged 6 to 10 months that will choose the helper toy is 0.5, is true.

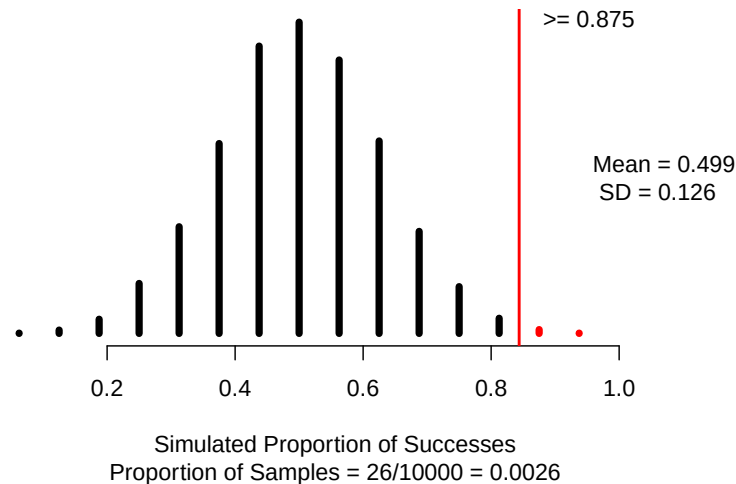
```
set.seed(100)
one_proportion_test(probability_success = 0.5, # Null hypothesis value
  sample_size = 16, # Enter sample size
  number_repetitions = 10000, # Enter number of simulations
  as_extreme_as = 0.875, # Observed statistic
  direction = "greater", # Specify direction of alternative hypothesis
  summary_measure = "proportion") # Reporting proportion or number of successes?
```



Notes on the null distribution

Using R, we will use the `one_proportion_test()` function (in the `catstats` package) to simulate another null distribution of sample proportions and compute a p-value. Using the provided R script file, fill in the values/words for each `xx` with your answers from question 1 in the one proportion test to create a null distribution with 10000 simulations. Then highlight and run lines 1–16. *Note that the value for the mean, standard deviation, and p-value changed slightly.*

```
set.seed(216)
one_proportion_test(probability_success = 0.5, # Null hypothesis value
  sample_size = 16, # Enter sample size
  number_repetitions = 10000, # Enter number of simulations
  as_extreme_as = 0.875, # Observed statistic
  direction = "greater", # Specify direction of alternative hypothesis
  summary_measure = "proportion") # Reporting proportion or number of successes?
```

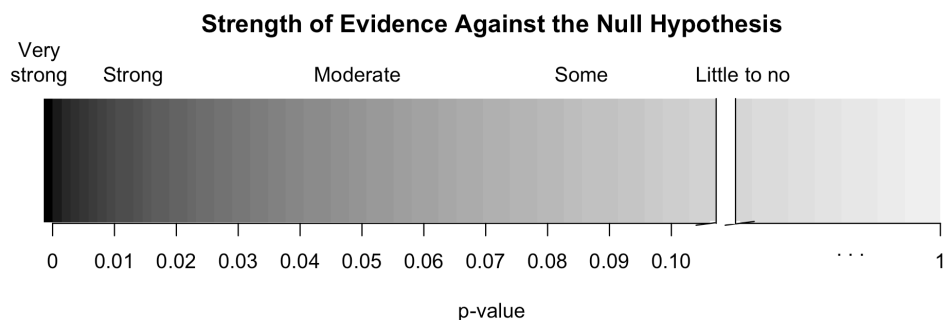


For the remainder of the questions in this activity use this 2nd simulated null distribution.

- Circle the observed statistic (value calculated in Activity 4) on the null distribution. Where does this statistic fall in the null distribution: Is it near the center of the distribution (near 0.5) or in one of the tails of the distribution?
- Is the observed statistic likely to happen or unlikely to happen if the true proportion of infants who choose the helper is 0.5? Explain your answer using the plot.
- Using the simulation, what is the proportion of simulated samples that generated a sample proportion at the observed statistic or greater, if the true proportion of infants who choose the helper is 0.5? *Hint: Look under the simulation.*

Notes on the p-value

The value in question 4 is the **p-value**. The smaller the p-value, the more evidence we have against the null hypothesis.



Interpret the p-value

The p-value measures the probability that we observe a sample proportion as extreme as what was seen in the data or more extreme (matching the direction of the H_A) IF the null hypothesis is true. This is a conditional probability, calculated dependent on the null hypothesis being true. Represented in probability notation:

$$P(\text{statistic or more extreme} | \text{the null hypothesis is true})$$

p-value interpretation:

Communicate the results and answer the research question

When we write a conclusion we answer the research question by stating how much evidence there is in support of the alternative hypothesis.

Conclusion:

Generalization

5. To what group of observational units can the results be generalized to?

3.4.4 Take-home messages

1. The null distribution is created based on the assumption the null hypothesis is true. We compare the sample statistic to the distribution to find the likelihood of observing this statistic.
2. The p-value measures the probability of observing the sample statistic or more extreme (in direction of the alternative hypothesis) if the null hypothesis is true.
3. The smaller the p-value of the test, the more evidence there is **against** the null hypothesis.

3.4.5 Additional notes

Activities 4–6 cover the material in Module 3. For more practice with simulation hypothesis testing of a single categorical variable use the Module 3 Review worksheet in the Unit 1 Review Materials (pgs. 112–115).

Use this space to summarize your thoughts and take additional notes on today's activity and material covered.

3.5 Activity 6: Helper-Hinderer — Simulation-based Confidence Interval

3.5.1 Learning outcomes

- Use bootstrapping to find a confidence interval for a single proportion.
- Interpret a confidence interval for a single proportion.

3.5.2 Terminology review

In today's activity, we will introduce simulation-based confidence intervals for a single proportion. Some terms covered in this activity are:

- Parameter of interest
- Bootstrapping
- Confidence interval

To review these concepts, see Chapters 10 & 14 in your textbook.

3.5.3 Helper-Hinderer

In the last class, we found very strong evidence that the true proportion of infants who will choose the helper character is greater than 0.5. But what *is* the true proportion of infants who will choose the helper character? We will use this same study to estimate this parameter of interest by creating a confidence interval.

As a reminder: A study by Hamblin, Wynn, and Bloom reported in Nature (Hamblin et al. 2007) was intended to check young kids' feelings about helpful and non-helpful behavior. Non-verbal infants ages 6 to 10 months were shown short videos with different shapes either helping or hindering the climber. Researchers were hoping to assess: Are non-verbal infants more likely to choose the helper toy? In the study, of the 16 infants age 6 to 10 months, 14 chose the *helper* toy and 2 chose the *hinderer* toy.

A **point estimate** (our observed statistic) provides a single plausible value for a parameter. However, a point estimate is rarely perfect; usually there is some error in the estimate. In addition to supplying a point estimate of a parameter, a next logical step would be to provide a plausible *range* of values for the parameter. This plausible range of values for the population parameter is called an **interval estimate** or **confidence interval**.

Activity intro

1. What is the value of the point estimate (sample statistic)?
2. If we took another random sample of 16 infants, would we get the exact same point estimate? Explain why or why not.

In today's activity, we will use bootstrapping to find a 95% confidence interval for π , the parameter of interest.

Notes on Confidence Intervals

Use statistical analysis methods to draw inferences from the data

3. Write out the parameter of interest in words, in context of the study. What does π represent?

To create the null distribution we flipped a coin 16 times to simulate infants randomly choosing the helper toy with a probability of 50%.

4. Why can't we use a coin to simulate the bootstrap distribution.

To create the bootstrap distribution.

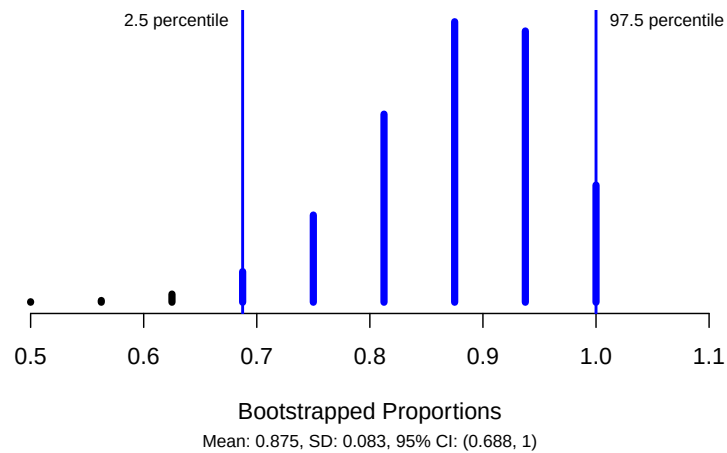
- First we would label the cards to represent the sample statistic: 14 helper and 2 hinderer.
 - Sample with replacement 16 times
5. Using the cards provided by your instructor, create one bootstrap sample. Report your simulated sample proportion on the whiteboard.

To use the computer simulation to create a bootstrap distribution, we will need to enter the

- “sample size” (the number of observational units or cases in the sample),
 - “number of successes” (the number of cases that choose the helper character),
 - “number of repetitions” (the number of samples to be generated), and
 - the “confidence level” (which level of confidence are we using to create the confidence interval).
6. What values should be entered for each of the following into the simulation to create the bootstrap distribution of sample proportions to find a 95% confidence interval?
 - Sample size (n):
 - Number of successes:
 - Number of repetitions (at least 10000):
 - Confidence level (as a decimal):

We will use the `one_proportion_bootstrap_CI()` function in R (in the `catstats` package) to simulate the bootstrap distribution of sample proportions and calculate a confidence interval. Using the provided R script file, fill in the values/words for each `xx` with your answers from question 6 in the one proportion bootstrap confidence interval (CI) code to create a bootstrap distribution with 10000 simulations. Then highlight and run lines 1–9.

```
one_proportion_bootstrap_CI(sample_size = 16, # Sample size
                             number_successes = 14, # Observed number of successes
                             number_repetitions = 10000, # Number of bootstrap samples to use
                             confidence_level = 0.95) # Confidence level as a decimal
```



Notes on the bootstrap distribution

95% Confidence Interval:

Interpretation of the 95% confidence interval in context.

Communicate the results and answer the research question

- Is the value 0.5 (the null value) in the 95% confidence interval?

Explain how this indicates that the p-value provides strong evidence against the null.

Statistical Analysis Write-up

In many of our studies we will write a paragraph summarizing the results of the study. The following is a summary paragraph for the infant study. The paragraph should include an introduction, the interpretation of the sample statistic, p-value and interpretation, confidence interval and interpretation, conclusion statement, and generalization.

An introduction including the observational units and variable(s) in the study and their role and type. The introduction should also include the sampling method and research question.

Researchers were interested if infants observe social cues and would be more likely to choose the helper toy. The observational units in this study are 16 non-verbal infants ages 6 to 10 months old. This is a convenience sample of infants. On each infant the researchers recorded whether or not the infant choose the helper toy. This is a categorical variable. Researchers were hoping to assess: Are non-verbal infants more likely to choose the helper toy?

Report and interpret the sample statistic in context of the study.

From the study, it was reported that the proportion of the 16 infants that chose the helper toy was 0.875.

Report how the p-value was found. Include an interpretation of the p-value.

A simulation null distribution with 10000 simulations was created in RStudio. The p-value was found by calculating the proportion of simulations in the null distribution at the sample statistic of 0.875 and greater. This resulted in a p-value of 0.0026. We would observe a sample proportion of 0.875 or greater with a probability of 0.0026, IF we assume the true proportion of non-verbal infants age 6 to 10 months who would choose the helper toy is 0.5.

Report and interpret the confidence interval.

In addition, a 95% confidence interval was found for the parameter of interest. We are 95% confident that the true proportion of non-verbal infants age 6 to 10 months who will choose the helper toy is between 0.688 and 1.

Write a conclusion for the study.

Based on the p-value of 0.0026, there is very strong evidence that the true proportion of non-verbal infants age 6 to 10 months who will choose the helper toy is greater than 0.5.

And last, report the scope of inference for the study. Since there is only a single categorical variable in this study, we only need to discuss generalization.

The results of this study can be generalized to the sample of non-verbal infants or similar observational units as the researchers did not select a random sample.

3.5.4 Take-home messages

1. The goal in a hypothesis test is to assess the strength of evidence for an effect, while the goal in creating a confidence interval is to determine how large the effect is. A **confidence interval** is a range of *plausible* values for the parameter of interest.
2. In simulation-based methods (bootstrapping), a simulated distribution of possible sample statistics is created showing the possible sample-to-sample variability. Then we find the middle X percent of the distribution around the sample statistic using the percentile method to give the range of values for the confidence interval. This shows us that we are $X\%$ confident that the parameter is within this range, where X represents the level of confidence.
3. When the null value is within the confidence interval, it is a plausible value for the parameter of interest; thus, we would find a larger p-value for a hypothesis test of that null value. Conversely, if the null value is NOT within the confidence interval, we would find a small p-value for the hypothesis test and strong evidence against this null hypothesis.
4. To create one simulated sample on the bootstrap distribution for a sample proportion, label n cards with the original responses. Draw with replacement n times. Calculate and plot the resampled proportion of successes.

3.5.5 Additional notes

Activities 4–6 cover the material in Module 3. For more practice with simulation confidence intervals for a single categorical variable use the Module 3 Review worksheet in the Unit 1 Review Materials (pgs. 112–115).

Use this space to summarize your thoughts and take additional notes on today's activity and material covered.

3.6 Golden Ticket to Descriptive and Inferential Statistical Methods

In this course, we have covered descriptive (summary statistics and plots) and inferential (hypothesis tests and confidence intervals) methods for five different scenarios:

- one categorical response variable (Module 3 & 4)
- one quantitative response variable (Module 6 & 7) or paired differences in a quantitative variable (Module 12)
- two categorical variables (Module 8 & 9)
- one quantitative response variable and one categorical explanatory variable (Module 11)
- two quantitative variables (Module 13)

The “golden ticket” shown on the next page presents a visual summary of the similarities and differences across these five scenarios.

Scenario	One Categorical Response	One Quantitative Response or Paired Differences	Two Categorical Variables	Quant. Response and Categ. Explanatory (independent samples)	Two Quantitative Variables
Type of plot	Bar plot	Dotplot, histogram, boxplot	Segmented bar plot, Mosaic plot	Side-by-side boxplots, Stacked dotplots or histograms	Scatterplot
Summary measure	Proportion	Mean or Mean difference	Difference in proportions	Difference in means	Slope or correlation
Parameter notation	π	μ or μ_d	$\pi_1 - \pi_2$	$\mu_1 - \mu_2$	β_1 or ρ
Statistic notation	$\hat{p}$	$\bar{y}$ or $\bar{y}_d$	$\hat{p}_1 - \hat{p}_2$	$\bar{y}_1 - \bar{y}_2$	b_1 or r
Null hypothesis	$H_0: \pi = \pi_0$	$H_0: \mu = \mu_0$ or $H_0: \mu_d = 0$	$H_0: \pi_1 - \pi_2 = 0$	$H_0: \mu_1 - \mu_2 = 0$	$H_0: \beta_1 = 0$ or $H_0: \rho = 0$
Conditions for simulation-based methods	Independent cases	Independent cases	Independent cases (within and between groups)	Independent cases (within and between groups)	Independent cases; Linear form
Simulation test (how to generate a null distn) p-value = proportion of null simulations at or beyond (H_A direction) the observed statistic	Spin spinner with probability equal to π_0 , n times or draw with replacement n times from a deck of cards created to reflect π_0 as probability of success. Plot the proportion of successes. Repeat 10000 times. Centered at π_0	Shift the original data by adding $(\mu_0 - \bar{y})$ or $(0 - \bar{y}_d)$. Sample with replacement from the shifted data n times. Plot sample mean or sample mean difference. Repeat 10000 times. Centered at μ_0 for a single quantitative response or 0 for paired data.	Label cards with response values from original data; mix cards together; shuffle into two new groups of sizes n_1 and n_2 . Plot difference in proportion of successes. Repeat 10000 times. Centered at 0.	Label cards with response variable values from original data; mix cards together; shuffle into two new groups of sizes n_1 and n_2 . Plot difference in means. Repeat 10000 times. Centered at 0.	Separate the (x,y) pairs. Hold the x values constant; shuffle new y 's to x 's. Find the regression line for shuffled data; plot the slope or the correlation for the shuffled data. Repeat 10000 times. Centered at 0.
Bootstrap CI (how to generate a boot. distn) X% CI: $\left(\frac{1-X}{2}\right)\%tile,$ $\left(X + \frac{1-X}{2}\right)\%tile)$	Label n cards with the original responses. Randomly draw with replacement n times. Plot the resampled proportion of successes. Repeat 10000 times. Centered at $\hat{p}$.	Label n cards with the original responses. Randomly draw with replacement n times. Plot the resampled mean difference. Repeat 10000 times. Centered at $\bar{y}$ for a single quantitative response or $\bar{y}_d$ for paired data.	Label n_1 cards with the original responses from group 1 and n_2 cards with the original responses from group 2. Keep groups separate. Randomly draw with replacement n_1 times from group 1 and n_2 times from group 2. Plot the resampled difference in proportion of successes. Repeat 10000 times. Centered at $\hat{p}_1 - \hat{p}_2$	Label n_1 cards with the original responses from group 1 and n_2 cards with the original responses from group 2. Keep groups separate. Randomly draw with replacement n_1 times from group 1 and n_2 times from group 2. Plot the resampled difference in means. Repeat 10000 times. Centered at $\bar{y}_1 - \bar{y}_2$.	Label n cards with the original (explanatory, response) pairs. Randomly draw with replacement n times. Plot the resampled slope or correlation. Repeat 10000 times. Centered at b_1 for slope or r for correlation.
Theory-based distribution	Standard Normal	t - distribution with $n - 1$ df	Standard Normal	t - distribution with min of $n_1 - 1$ or $n_2 - 1$ df	t - distribution with $n - 2$ df
Conditions for theory-based hypothesis tests and confidence intervals	Independent cases; Number of successes and number of failures in the sample both at least 10.	Independent cases; $n < 30$ with no clear outliers OR $30 \leq n < 100$ with no extreme outliers OR $n \geq 100$	Independence (within and between groups); Number of successes and number of failures in EACH sample all at least 10. (All four cell counts at least 10.)	Independent cases (within and between groups); In each sample, $n < 30$ with no clear outliers OR $30 \leq n < 100$ with no extreme outliers OR $n \geq 100$	Linear form; Independent cases; Nearly normal residuals; Variability around the regression line is roughly constant.
Theory-based standardized statistic (test statistic)	$Z = \frac{\hat{p} - \pi_0}{SE_0(\hat{p})}$ $SE_0(\hat{p}) = \sqrt{\frac{\pi_0 \times (1 - \pi_0)}{n}}$	$T = \frac{\bar{y} - \mu_0}{SE(\bar{y})}$ OR $T = \frac{\bar{y}_d - 0}{SE(\bar{y}_d)}$ $SE(\bar{y}) = \frac{s}{\sqrt{n}}, SE(\bar{y}_d) = \frac{s_d}{\sqrt{n}}$	$Z = \frac{\hat{p}_1 - \hat{p}_2 - 0}{SE_0(\hat{p}_1 - \hat{p}_2)}$ $SE_0(\hat{p}_1 - \hat{p}_2) = \sqrt{\widehat{p_{pool}} \times (1 - \widehat{p_{pool}}) \times \left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$	$T = \frac{\bar{y}_1 - \bar{y}_2 - 0}{SE(\bar{y}_1 - \bar{y}_2)}$ $SE(\bar{y}_1 - \bar{y}_2) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$	$T = \frac{b_1 - 0}{SE(b_1)}$ $SE(b_1)$ is the reported standard error (std. error) of the slope term in the lm() output from R.
Theory-based confidence interval	$\hat{p} \pm z^* \times SE(\hat{p})$ $SE(\hat{p}) = \sqrt{\frac{\hat{p} \times (1 - \hat{p})}{n}}$	$\bar{y} \pm t^* \times SE(\bar{y})$ $\bar{y}_d \pm t^* \times SE(\bar{y}_d)$ $SE(\bar{y}) = \frac{s}{\sqrt{n}}, SE(\bar{y}_d) = \frac{s_d}{\sqrt{n}}$	$\hat{p}_1 - \hat{p}_2 \pm z^* \times SE(\hat{p}_1 - \hat{p}_2)$ $SE(\hat{p}_1 - \hat{p}_2) = \sqrt{\frac{\hat{p}_1 \times (1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2 \times (1 - \hat{p}_2)}{n_2}}$	$\bar{y}_1 - \bar{y}_2 \pm t^* \times SE(\bar{y}_1 - \bar{y}_2)$ $SE(\bar{y}_1 - \bar{y}_2) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$	$b_1 \pm t^* \times SE(b_1)$ $SE(b_1)$ is the reported standard error (std. error) of the slope term in the lm() output from R.

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